INTENT BEFORE ACTION: DECODABLE AND PORTABLE SUB-GOAL REPRESENTATIONS IN LLMS

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ABSTRACT

When a Large Language Model (LLM) writes code to satisfy a multi-step specification, does it know *what it intends to do next* before it starts writing? We show that code LLMs encode a compact representation of their next instrumental sub-goal before emitting its implementation, and that this encoding is *decodable* and *portable*. Using causal mediation analysis, we localize this encoding to a set of attention heads, which we call the *goal heads*. We introduce GOALSCOPE, a lightweight probe that uses what these heads write to the residual stream to verbalize the upcoming sub-task in natural language. We demonstrate that the encoding is abstract: it can be extracted from one context and transplanted to trigger the same sub-goal in a different context and programming language. These heads are a crucial bottleneck in how the LM communicates the intent encoding to later layers for its concrete realization: preserving their access to task information largely preserves the capability of executing the next sub-task, while restricting that access in a broader set of goal heads sharply impairs it. We replicate the readout and transfer findings across LLM families and sizes, and explore the same heads' readouts in non-code settings. Our results provide concrete, testable evidence that LLMs encode abstract representations of the next instrumental goal before acting on it.

1 INTRODUCTION

Consider a programmer asked to implement *merge sort*, a task that decomposes into several sub-tasks: *split the array*, *sort each half*, and *merge the sorted halves*. Before writing the code for the final step, the programmer already has a coherent goal: *"merge the two halves."* We ask whether large language

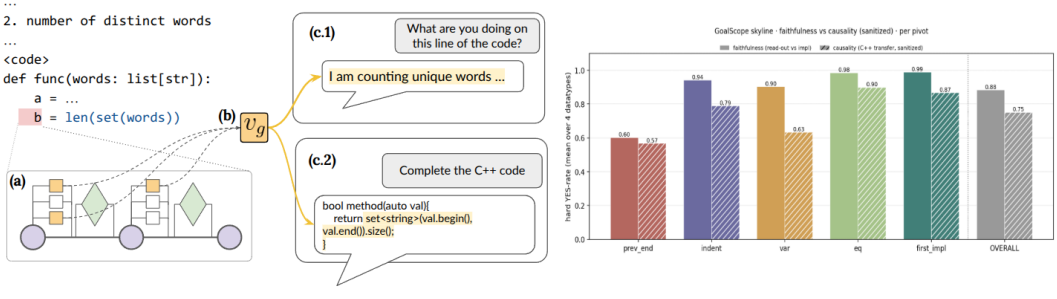


Figure 1: **The GoalScope readout setup.** (a) At the highlighted pivot (" b"), we cache the selected attention heads' outputs that are added to the residual stream. (b) We aggregate these outputs into a single vector v_g. (c) In a separate LM run, the user may ask different queries, such as (c.1) asking the LM to verbalize the next step, (c.2) implement the next step in a different programming language. We steer the LM response by adding v_g at each generation position. The query can be adapted to elicit different aspects of the model's encoding of its context and upcoming behavior. The right panel shows that we can get decent readouts as early as the indent (" ") token, 3 tokens before the concrete implementation begins.

lisa: "abstract" vs "portable" - on second thought, if you define abstract, it's fine to leave it :)

natalie: What is the definition of "intent"? It is loaded. I'm concerned that "intent" makes a stronger psychological claim than our evidence supports. It implies not only a plan pursuing a goal but also hints at a subjective experience. Consider, for example, a billiard ball moving toward another ball following the laws of physics. Do you say that the ball *intends* to strike the other ball, or do you say that it is *going* to strike it? In this sense, when we relate intentions for LLMs, we are overclaiming, as we have no way—nor is any known—to measure that subjective experience. My proposal: replace the word "intent" with "plan" or "prediction".

lisa: suggestion: does it plan each step before outputting/generating an implementation

lisa: could drop "a compact representation of" as it's already covered by "encode"

lisa: alt suggestion: We introduce GOALSCOPE, a lightweight probe trained on the residual stream to ...

lisa: minor comment: i'd stay consistent between embedding and representation.

lisa: looks great!

arnab: I am using terms like "sub-goal", "intention", ... interchangeably in the intro. We need to come to a consensus on what term to use.

lisa: I like subgoal/goal

models (LLMs) write code the same way: when an LLM emits a long span of tokens to satisfy a multi-step specification, does it hold an explicit representation of the *next* instrumental sub-goal before it begins its implementation?

lisa: could we use planning instead of instrumental? otherwise maybe we should use instrumental in the abstract

We study the *instrumental* level of this hierarchy: the *broader goal* (implement merge sort) decomposes into *instrumental sub-goals* (merge the two sorted arrays), each of which admits many *implementations* (such as using a *for* loop or a *while* loop). Our question concerns a representation of *which sub-goal the model is about to pursue*, and whether that representation can be separated from its concrete realizations in code.

Although LLMs are trained to predict one token at a time, a growing body of work shows that they *look ahead*: they encode information about tokens they have not yet emitted (Pal et al., 2023) and plan toward constraints several tokens in the future (Lindsey et al., 2025). We ask what the plan for the *next step* looks like as a concrete internal object, and whether that object is one we can read and move between contexts.

arnab: maybe move to Related Works if the intro gets long.

We investigate this question through activation patching on code tasks with multiple sub-goals. Our main experiments use `Qwen2.5-Coder-32B-Instruct`, with replications across model families and sizes. Our analysis yields three key findings.

lisa: this could also be moved up to support the context?

The next sub-goal is encoded before implementation, and is verbalizable (Section 3). We localize the encoding of the LLM's next sub-goal to a compact set of attention heads, which we call the *goal heads*. We introduce GOALSCOPE, a lightweight probe built on these heads that verbalizes the upcoming sub-goal directly from the LM's internal activations. Before the model writes the implementation, we can extract what the heads write to the residual stream and use it to generate a faithful description of the operation it is about to perform.

The encoding is abstract and portable (Section 3.2). Transplanting the encoding steers a fresh code context to pursue the source sub-goal. A goal extracted from Python can trigger the same operation in C++, and vice versa. This portability provides evidence that the heads carry a representation of the operation that generalizes across its surface realizations. We also explore whether the same heads can verbalize sub-goals in code whose component steps are not explicitly listed, and inferred relations in non-code few-shot tasks (Section 6).

Goal heads mediate the model's implementation of the sub-goal (Sections 4 and 5). A small set of heads suffices for effective readout, and a broader set is causally important for carrying out the operation. When we restrict other upper-layer heads' access to the upcoming task and code span, preserving goal-head access largely preserves implementation performance. Conversely, restricting the broader goal-head set to the attention sink sharply reduces performance, while restricting random heads has little effect. The heads attend to relevant task descriptions, and changing their query or key/value states can redirect the sub-goal pursued. Using their value and output projections, we also introduce **OVReader** to trace where the upcoming sub-goal becomes readable in intermediate residual states.

Our results provide concrete, testable evidence that code LLMs represent and pursue intermediate goals: a compact, portable, causally manipulable encoding of *intent* that precedes action.

arnab: Teaser figure: the readout + one steering example, framed as "intent before action".

2 TASK AND INTERVENTION SETUP

2.1 LANGUAGE MODELS AND HEAD CONTRIBUTIONS

An autoregressive transformer $M : \mathcal{X} \to \mathcal{Y}$, over a vocabulary $\mathcal{V}$ maps a sequence of tokens $x = (x_1, \ldots, x_n)$, $x \in \mathcal{V}$, to $y \in \mathbb{R}^{|\mathcal{V}|}$, a probability distribution over its next token. Internally, M has L layers; the residual stream at layer ℓ and position t is $h_t^\ell = h_t^{\ell-1} + m_t^\ell + \sum_j a_t^{\ell j}$, where m_t^ℓ is the output of the MLP and $a_t^{\ell j}$ is the contribution of the j-th attention head. Suppressing normalization and positional encoding, a head's contribution to the residual stream is

$$a_t^{\ell j} = \left(\sum_{i \le t} \mathrm{Attn}(q_t, K)_i v_i \right) W_O^{\ell j}, \qquad \mathrm{Attn}(q_t, K) = \mathrm{softmax}\left(\frac{q_t K^\top}{\sqrt{d_{\mathrm{head}}}} \right). \tag{1}$$

with query $q_t = h_t^{\ell-1} W_Q^{\ell j}$, keys $K = h_{\leq t}^{\ell-1} W_K^{\ell j}$, affinity of query and key states determine the mixture over the value states $v_i = h_i^{\ell-1} W_V^{\ell j}$, which is then projected with the output projection $W_O^{\ell j}$ and added to the residual stream. In grouped-query attention, multiple query heads read from the same key/value head.

2.2 Sequential subgoal implementation

We study a set of controlled settings in which the model must satisfy a *broader goal* that decomposes into an ordered set of independent *sub-tasks* $G = \{g_1, \ldots, g_K\}$, each specified in natural language (for example, *"check if there are any palindromes in a list of strings"*). A prompt p presents the specification and asks the model to implement the sub-tasks in code, one after another. The model realizes each g_k over a contiguous span of generated tokens. We use two terms throughout. The *pivot* π_k is the token position at which the model is about to begin sub-task g_k, which we dub as the *frontier*. In our controlled setup, we prefill in the LM's progress until the *pivot* π_k. This let's us ask if the upcoming assignment is already encoded in the *pivot* π_k and compare the readout with the LM's actual *future* behavior. A concrete example of the task is shown on Figure 1, with more details on Appendix A.

Our main evaluation contains 253 specifications over integers, strings, lists of integers, and lists of strings. Each prompt lists two to four operations, which we refer to as predicates, including Boolean tests, counts, and other transformations. We inspect five positions around the frontier: the end of the preceding line, indentation, the variable name, the assignment operator, and the first implementation token. The first four are pre-implementation capture positions, which we call *pivots*; the fifth provides a comparison after implementation has begun. We use the assignment operator as the principal pivot for the ablation experiments. The corpus supplies 1,265 observations per evaluation mode, with five repeated measurements per specification. The data distribution, complete example prompt, and tokenization are in Appendix A and **??**.

arnab: condense the different pivot explanation to figure. nothing in text. stash specifics in the appendix.

2.3 Intervention Primitives

Activation patching (Vig et al., 2020; Mueller et al., 2026). We run M on a prompt p but overwrite a chosen set of internal activations with latents cached from a different run, and read out the effect on the output distribution. We write $M(p)\left[a_i^{\ell j} \leftarrow v\right]$ for the forward pass on p in which the output of head $[\ell, j]$ at position i is replaced by the vector v. When the replaced value is cached from a *source* prompt p_{src} and injected into a *destination* prompt p_{dst}, a change in $M(p)\left[v_{\text{orig}} \leftarrow v_{\text{rep}}\right]$ isolates the causal role of the patched component/encoding.

arnab: M(x) usually denotes a single forward pass. Our setup is multi-token generation. Ok?

Query, key/value, and output transplants. The head decomposition in Eq. 1 exposes three places a transplant can act, and we use all three. An *output* (OV) transplant replaces a head's contribution $a_t^{\ell j}$ (what it *writes* to the residual stream), and is the primitive behind GoalScope and our head localization (Sections 3 and 4). A *query* transplant replaces the query $q_t^{\ell j}$ that the destination uses at π_k with the source's, changing what the read *looks for*. A *key/value* transplant replaces the keys and values $K^{\ell j}, V^{\ell j}$ at the specification tokens with the source's, changing what the read *finds*. We use these transplants both to *steer* a fresh context toward an intended sub-goal and to probe *where* the goal is computed (Section 5).

3 The Next Sub-Goal Is Encoded Before Implementation

Can we read what the model is about to do before it writes the implementation? We first locate attention heads that carry a portable representation of a code operation. We then use their contributions at the pivot to describe the upcoming sub-goal and to trigger its implementation in a new context.

3.1 Localizing goal heads

We begin by locating a set of attention heads (*goal heads*) that carry the encoding of the next sub-goal on a simpler, single-predicate task. A source and a destination prompt request different predicates on the same datatype, written in different programming languages. Both prompts end where the next token distinguishes the requested implementations. We cache the source head contributions and

learn a sparse mask $m^{\ell j}$ jointly over attention heads. In the destination run, the mask controls which contributions are replaced:

$$a_{-1}^{\ell j} \leftarrow \text{mask}^{\ell j} a_{\text{src}}^{\ell j} + (1 - \text{mask}^{\ell j}) a_{-1}^{\ell j}. \tag{2}$$

The mask is optimized to promote a token that would begin implementing the *source predicate in the destination language*. We choose this target to differ from the source's next token, so the localization objective rewards transfer across different token realizations. We call the selected set H the *goal heads*. Our main selection contains 122 heads in layers 28–55 of the 64-layer model. The mask-learning procedure and head-set construction are described in Appendix B. We next test the selected heads on full readout continuations.

3.2 GoalScope and implementation transfer

We use the attention heads identified in 3.1 to build GoalScope, a lightweight probe that verbalizes the sub-goal encoded at a single token position. Building on Patchscope (Ghandeharioun et al., 2024), we cache what the selected heads write to the residual stream at the source pivot and sum these contributions into one vector. A separate run of the same model receives a fixed prompt asking what operation it is performing. We inject the cached vector into this probe's residual stream and generate a natural-language description. The source task reaches the probe through the injected activations; its instructions and code are not supplied as text.

Our main configuration adds the vector at layers 35–45, scaled to 0.1 times the local residual norm. We reuse the same source vector at the probe's last prompt position and every subsequent decoding step, generating the readout greedily. Appendix C gives the exact prompt, normalization, and other design choices to perform the intervention.

Faithfulness. We ask whether the verbalization faithfully describes a nontrivial part of what the LM actually generates as its implementation of the next *meaningful* step. In our controlled setup, we prefill until a target sub-task g_k, giving us a gold target to measure the fidelity of the readouts.

Causality. We inject the same encoding into a code stub in another programming language and ask whether it triggers a nontrivial part of the source operation g_k. For example, an encoding captured from Python is used to steer a C++ continuation. This tests whether the encoding of g_k, captured by the *goal heads*, are abstract from the concrete tokens that implement it and whether this encoding can nudge the LM towards implementing the same sub-task g_k in a new context (C++) with a different sequence of generated tokens.

These two measures expose complementary properties of the encoding: a natural-language description reveals its semantic content, while cross-language transfer shows that it can guide a different realization of the operation. For both measures, we ask an LLM oracle (`claude-sonnet-4-6`) to compare the readout with the extracted LM implementation of the frontier g_k on the source prompt. A positive judgment credits nontrivial agreement, including partial implementations. For *causality* we ensure that the frontier is not aligned with the LM's default behavior to the specific prompt. Exact oracle prompts with additional details are in **??**.

3.3 Readout across capture positions

arnab: doesn't need a different section. illustration of pivot positions + results should be absorbed in **??**.

We next examine how compactly the encoding can be read and how the heads contribute to the model's own implementation.

4 Goal Heads Carry and Mediate the Sub-Goal

How many heads are needed to read the sub-goal, and how important are these heads for carrying it out? We study both questions: first by varying the number of heads used for readout, then by restricting their access to task information during the model's own code generation.

arnab: Figure 2 and Figure 2 needs to be side-by-side. save space on the main text. maybe Figure 2 should be a bar-chart.

4.1 Small head sets support effective readout

Twenty-five selected heads, approximately 0.98% of the model's 2,560 query heads, achieve 93.7% readout faithfulness and 90.5% implementation transfer at the assignment pivot. The 122-head

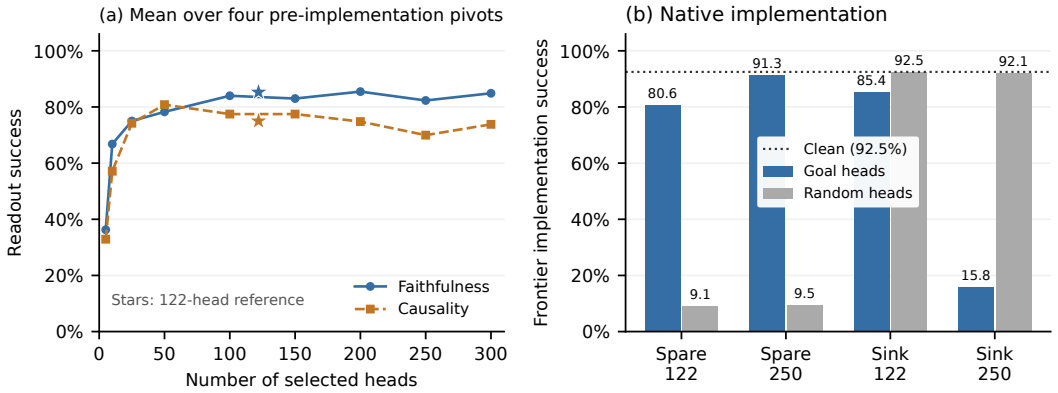


Figure 2: **Compact readout and causal involvement of goal heads.** Left: faithfulness and causality versus head budget, averaged over the four pre-implementation pivots ($n = 1,012$, $\alpha = .1$); stars mark the separately selected 122-head reference. Selections are not strictly nested. Right: native frontier implementation success on the same specifications. "Spare" preserves access to the frontier task/code span for the selected heads while blocking other heads in layers 20–63; "sink" restricts the selected heads to key position zero. Random controls receive the corresponding intervention. The dashed line is clean performance (.925). Exact rates and intervention diagrams appear in Appendix F; individual pre-implementation pivots appear in Appendix D.2.

reference achieves 97.2% and 95.7%, respectively (Figure 2). Much of the sub-goal's semantic content can therefore be recovered from a small fraction of the attention heads.

arnab: recheck the numbers.

Larger selections do not improve performance monotonically. The selections differ in membership and are not strictly nested, so the curves compare available head sets rather than isolate head count alone.

4.2 Goal heads mediate implementation

We next ask whether the model relies on the *goal heads* to implement the next sub-goal. On the same 253 specifications, we intervene at the assignment pivot and continue the intervention during decoding. We compare the original generation with interventions on the 122-head core, a broader 250-head set, and random-head controls.

Sufficiency of goal-head access. We preserve the selected heads' access to the upcoming task and code span while blocking that access for other heads in layers 20 and above. The blocked span begins just before the frontier description and ends at the final prompt token. Lower layers, MLPs, and attention to positions outside this span operate as usual. This tests whether goal-head access can sustain implementation when other upper-layer attention heads lose access to the same information. The control preserves access for a comparable number of random heads distributed on the upper layers. The exact spans and sampling pools are in Appendix F.

arnab: needs illustration of the ablation setup with a specific prompt, either here or in the appendix

Necessity of goal-head access. In the complementary experiment, we restrict the selected heads to attend only to position zero, the attention sink, while other heads operate normally. The selected heads still contribute the sink value but lose access to the task and code tokens. We compare with applying the same restriction to an equal-size random set, drawn from the depth pool and disjoint from the goal heads.

Figure 2 shows that the LM's competency of implementing the the next sub-goal is largely preserved when only goal heads can access the task information, with most of the heads in deeper layers are blinded. This indicates that the smaller set of 122 goal heads are sufficient in mediating the g_k encoding to later layers for concrete implementation. On the complementary necessity experiment we find that blinding a broader set of 250 goal heads reduces LM competency sharply, confirming that the goal heads are a crucial bottleneck mediating the g_k encoding to the implementation. However, blinding the smaller set of 122 heads keeps the performance largely intact. This means the LM is

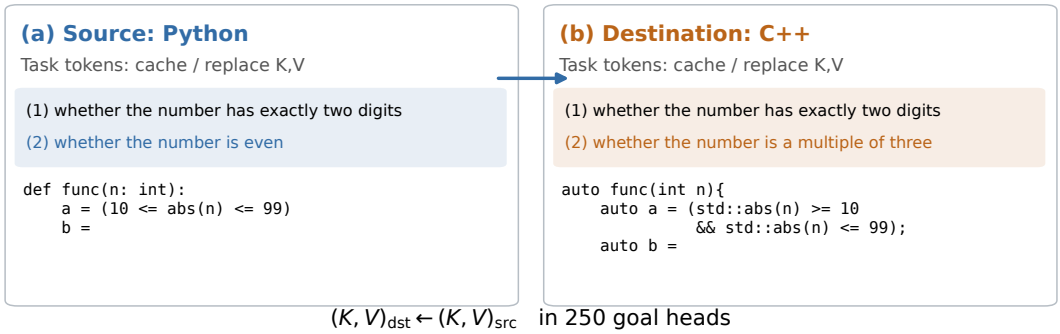


Figure 3: **Task-span KV transfer changes the next operation.** A saved Python-to-C++ example: both prompts have already implemented the shared two-digit check. We replace the keys and values of the destination task span in 250 goal heads with those from the source, whose second task is evenness. The destination still requests divisibility by three, but its patched continuation checks divisibility by two. GoalScope's readout also describes evenness. Chat wrappers and later return statements are omitted; the long C++ prefill line is wrapped for display. The patch is active across prefill and decoding, with no non-goal-head ablation and selected-head OV amplification of 2.

Source \ Destination	Python	C++
Python	99.2 {98.4}	98.4 {93.8}
C++	99.2 {96.1}	100.0 {96.1}

Table 1: **KV transfer across programming languages.** Rows are source languages and columns destination languages. Each cell reports causal success {source readout}, in percent, across four datatypes ($n = 128$ per cell; 512 total). Causal success counts source-only or both-goal verdicts; the readout score counts an exclusively source-directed patched GoalScope description. Both use all examples in the denominator. Neither score certifies full program correctness. Datatype breakdowns and the baseline-valid shift criterion are in Appendix H.

flexible enough to compensate for the ablation by using other *goal heads* (not in the 122 set) to communicate g_k to later layers.

5 How Goal Heads Access the Upcoming Sub-Goal

What information do the *goal heads* use to guide the next operation? We examine where the heads attend, how changing their queries or task-span keys and values affects the continuation, and where the sub-goal can be read from the residual stream.

5.1 Attention to Relevant Task Descriptions

The *goal heads* often look back to the task they are about to implement. At a frontier position, their collective value-weighted attention concentrates on tokens in the prompt describing the upcoming predicate. This resembles the *gaze* behavior observed during generation in other settings (Gandikota & Bau, 2026). These qualitative attention patterns suggest a connection between the task descriptions and the goal encoding. We test that connection causally by changing the information available through the heads' keys and values.

5.2 Key-Value Transfer Overwrites the Frontier

arnab: simplified figure showing the KV-transfer.

Consider a source prompt with tasks (A, B) and a destination with tasks (A, D), both with A already implemented. The next requested sub-goal is B in the source and C in the destination. We cache the keys and values of the source task tokens and transplant them into the destination's goal-head attention. The destination prompt still requests C; the patched heads now access the source task information B. We ask whether this change makes the model implement B.

arnab: needs concrete illustrative example.

In Figure/Table X, we show 2 scores to evaluate the setup: (i) *causal-shift*, whether the LM implements the source frontier; and (ii) *readout-shift*, whether the GOALSCOPE shifts to the source frontier. This KV-transfer experiment with 250 goal heads show strong generalization across datatypes, even cross-language transfer. The success rate of the KV-transfer in overwriting the destination frontier complements Section 4.2's finding that the goal head set are a crucial bottleneck in the LM's mediation of the sub-goal signal to later layers. The cross-language intervention success also confirm that the goal heads *read* the task information to encode an abstract representation of the next sub-task g_k, which when patched to a different environment is realized in a different sequence of tokens.

arnab: show the concrete scores, across language confusion table with the format `causality {read-out shift}` here.

5.3 QUERIES ALREADY KNOW THE NEXT SUB-GOAL

arnab: not yet sure if this should be in the main text

5.4 READING INTERMEDIATE RESIDUAL STATES WITH OVREADER

arnab: no need to show 2 figures on Figure 4. just one figure showing both faithfulness and causality lines together, with standard deviation across the datatypes. maybe pair the figure with the 122 goal head distribution in the model.

At what depth does the upcoming sub-goal become readable? We introduce **OVReader**, which uses the goal heads' value and output projections to read intermediate residual states. Formally, an attention head, its output and value projection read from $OV(h) = W_O W_V \ln(h) = {}^{1}/_{rms(h)} \times W_O W_v(\mu \odot h)$. Qwen-2.5 series of LMs use $\mathrm{RMSNorm}$ with μ as the element-wise stretching and ${}^{1}/_{rms(h)}$ is the rescaling to the unit ball. If we ignore the rescaling the OV read becomes, $OV(h) = W_O W_V \mathrm{diag}(\mu)\ h$. $W_O W_V \mathrm{diag}(\mu)$ can be absorbed into a single transformation W_{OV}. For multiple goal heads we can absorb their OV reads into a single transformation $\mathbf{W_{OV}}$, by simply adding each head's OV-transformation matrices.

arnab: add citations to sheridan's work. rename the terms to match exactly. polish the presentation

We apply the aggregated transformation $\mathbf{W_{OV}}$ on a residual state h, and use the output as the steering vector (on Figure X) to get the readout, which we score using the same *faithfulness* and *causality* scores introduced in Section 3.2. Figure 4 shows the readout performance across layers on the " =" token, right before the LM's g_k implementation starts. OVReader readout performance already peaks by layer 40, before the layers where most of the goal heads are concentrated. So, by the time most of the goal heads take action, the next sub-task g_k is already committed and is decodable in the residual latents.

arnab: maybe Appendix is a better place for the full derivation.

arnab: needs an end.

6 BEYOND EXPLICITLY SPECIFIED CODE TASKS

arnab: didn't check any writing here. will depend on the concrete findings from ongoing experiments.

Do the heads localized on code also carry an operation that the model infers from examples? Our controlled code tasks name each sub-goal in the prompt. We now examine a few-shot setting where the intended relation is conveyed through demonstrations.

Few-shot relation tasks. Following the function-vector setting (Todd et al., 2024), we provide input–output pairs demonstrating a relation, followed by a new input. We retain the 122 heads localized on code and adapt the probe question to ask for the hidden rule in a short phrase. For example, an antonym prompt contains the following pairs:

```
Q: enter  A: exit
Q: feasible  A: infeasible
. . .
Q: mercy  A:
```

The model answers "cruelty" and continues generating question–answer pairs. At the newline following "cruelty," GOALSCOPE reads *"Identifying antonyms."* At the answer-prefix token `A` before a later answer, "slow," it again reads *"Identifying antonyms."* The probe is given neither the name of the relation nor the demonstrations as text. The code-localized head contributions carry enough information to describe the inferred operation.

These are qualitative readouts during the continuing sequence, with the first captured after the initial answer and the second before a later answer. Nearby positions also produce vague or incorrect

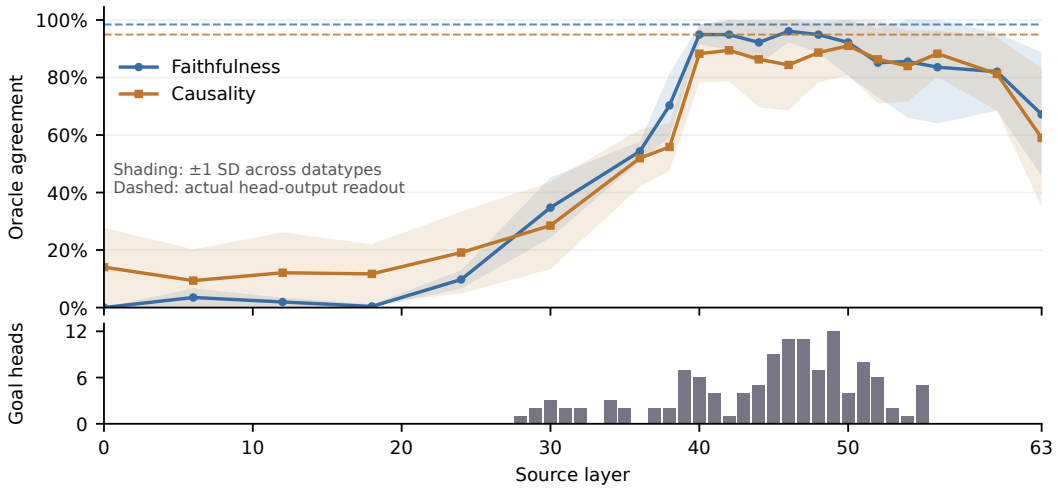


Figure 4: **OVReader at the assignment pivot.** Faithfulness and causality are averaged over four datatypes, with shaded bands showing ± 1 sample standard deviation across datatype rates (64 examples per datatype). Dashed lines show readout from the selected heads' actual outputs on this same cohort. Below: the distribution of the 122 goal heads by layer. These archived 256 examples form a separate cohort from **??**; OVReader uses $\alpha = .15$ and the head-output reference uses .1. Causality scores include trivial-readout sanitization. Other capture positions and the evaluated per-head normalization are given in Appendix I.

descriptions. Appendix M gives the full demonstration list, exact probe wording, token indices, and contrasting readouts. The example shows that the code-localized heads can expose an inferred non-code relation; a quantitative evaluation across relations and positions remains to be completed.

Code without enumerated sub-goals. Free-form code poses a related question: can the readout describe a step that the model chooses while solving a broader programming request? Here, evaluation must identify a meaningful operation in the model's continuation, which may be a local statement or a larger enclosing computation. This extension requires qualitative examples and an explicit choice of sub-goal granularity.

We report replications within other model families and sizes in Appendix K. Further exploratory non-code settings are discussed in Appendix M.

7 Related Work

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Planning and look-ahead in LMs. Prior work shows that autoregressive LMs encode information about future tokens (Pal et al., 2023) and plan toward constraints in later output (Lindsey et al., 2025). We study the upcoming computation at the level of an instrumental sub-goal. Our experiments ask whether its representation can be read before implementation and transferred to guide the same operation in a new context.

claude: Add citations: future-lens / prompt-injection look-ahead, poetry-planning (Anthropic biology), chess/maze planning, emergent world models.

Portable task representations and function abstractions. Function vectors are compact representations of demonstrated input–output tasks, carried by a small set of attention heads and transferable to new contexts (Todd et al., 2024). Concept-induction heads carry word representations that mediate translation across languages (Feucht et al., 2025). Filter heads encode predicates in their query states and reapply them across collections, formats, languages, and tasks (Sen Sharma et al., 2026). Our work examines a related abstraction during multi-step generation: the particular sub-goal the model is about to implement. We localize its encoding in head contributions, verbalize its content, and test its portability across programming languages.

Verbalizing internal representations. Patchscope uses a language model to express information extracted from its hidden states (Ghandeharioun et al., 2024). GoalScope applies this approach to

the contributions of causally localized attention heads at a single source position. The cached signal is reused throughout probe generation to obtain a multi-token description or code implementation. OVReader extends this readout to intermediate residual states using the selected heads' value and output projections.

claude: Add: logit lens, tuned lens, SelfIE, function-vector readouts; contrast what GOALSCOPE reads (head-output, multi-token) vs. residual-stream single-token lenses.

High-level interpretability of goal-directed behavior. The high-level interpretability agenda studies model behavior through abstractions that can be investigated without a complete account of every underlying computation (Greenblatt, 2024). We use an instrumental sub-goal as such an abstraction. Cross-language transfer tests whether its encoding can guide different implementations, while restrictions on goal-head attention test its role in the model's own generation.

claude: Cite causal abstraction / DAS (Geiger et al.), high-level causal mediation; position our cross-language transfer as the abstraction test.

8 DISCUSSION

We find that code LLMs carry a compact representation of their upcoming sub-goal before writing its implementation. The representation can be extracted from attention-head contributions, verbalized in natural language, and transplanted to trigger the same operation in another programming language. The heads also mediate the model's own implementation: preserving their access to task information largely preserves performance, and restricting a broader set sharply reduces it. These findings connect an interpretable description of what the model is about to do with an internal representation that can influence its behavior.

arnab: didn't check anything. will depend on the concrete framing and findings.

An abstraction of the upcoming operation. The portability of the encoding suggests a separation between the identity of a sub-goal and its realization in code. An operation extracted from Python can guide a C++ implementation, even though the two require different tokens. Natural-language verbalization provides a complementary way to inspect this semantic content. Our evidence concerns the upcoming operation near the start of its implementation; it leaves open how much of a longer plan is represented and how its details are chosen. Verbalizability also does not establish that the representation's native format is natural language.

From goal representation to goal selection. The intervention results identify several parts of goal-directed computation. Queries carry information that can favor a particular sub-goal; task-span keys and values supply information to which the model can adapt; and the heads' projections expose the upcoming operation in intermediate residual states. The ablations further distinguish a small set sufficient for readout from a broader set important for implementation. These observations constrain the mechanism, while leaving open how the model combines completed work with the specification to select its next step. Independently varying progress, task order, and target identity offers a way to investigate that computation. The current interventions establish causal involvement under specific access restrictions, rather than a unique bottleneck or an isolated goal-selection circuit.

Scope and limitations. Our principal quantitative setting uses explicit task specifications and a known frontier. The oracle measures nontrivial semantic agreement and can credit partial implementations, so success rates should be interpreted separately from executable program correctness. Capture positions are repeated observations of the same specifications, and head-budget comparisons use selections that are not strictly nested. Replications use each model's own heads and activations; differences between models combine checkpoint, head-selection, and generation effects. The few-shot example is qualitative, and evaluating free-form code requires further examples and decisions about sub-goal granularity. These settings are promising tests of how far the code-localized representation generalizes.

GOALSCOPE provides a concrete tool for examining what a model is about to do as generation proceeds. Further work can use these readouts to study how sub-goals change, when an intended operation fails to be implemented, and how the model decides to move on to the next step. Our exploratory boundary-signal interventions (Appendix N) provide a complementary starting point for the last question.

ETHICS

REPRODUCIBILITY

ACKNOWLEDGEMENTS

REFERENCES

Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, et al. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 1(1):12, 2021.

Sheridan Feucht, Eric Todd, Byron C Wallace, and David Bau. The dual-route model of induction. In *Conference on Language Models (COLM)*, 2025. URL https://openreview.net/forum?id=bNTrKqqnG9.

Rohit Gandikota and David Bau. Gaze heads: How vlms look at what they describe. *arXiv preprint arXiv:2606.14703*, 2026.

Asma Ghandeharioun, Avi Caciularu, Adam Pearce, Lucas Dixon, and Mor Geva. Patchscopes: A unifying framework for inspecting hidden representations of language models. In *International Conference on Machine Learning (ICML)*, 2024.

Ryan Greenblatt. High-level interpretability: how LLMs make decisions in pursuit of goals, 2024. Interview remarks; high-level, intervention-style interpretability of goal-directed behavior.

Jack Lindsey, Wes Gurnee, Emmanuel Ameisen, Brian Chen, Adam Pearce, Nicholas L. Turner, Craig Citro, Joshua Batson, et al. On the biology of a large language model. *Transformer Circuits Thread*, 2025. URL https://transformer-circuits.pub/2025/attribution-graphs/biology.html.

Aaron Mueller, Jannik Brinkmann, Millicent Li, Samuel Marks, Koyena Pal, Nikhil Prakash, Can Rager, Aruna Sankaranarayanan, Arnab Sen Sharma, Jiuding Sun, et al. The quest for the right mediator: Surveying mechanistic interpretability for nlp through the lens of causal mediation analysis. *Computational Linguistics*, 52(1):331–378, 2026.

Koyena Pal, Jiuding Sun, Andrew Yuan, Byron C. Wallace, and David Bau. Future lens: Anticipating subsequent tokens from a single hidden state. In *Conference on Computational Natural Language Learning (CoNLL)*, 2023.

Arnab Sen Sharma, Giordano Rogers, Natalie Shapira, and David Bau. LLMs process lists with general filter heads. In *International Conference on Learning Representations (ICLR)*, 2026. URL https://filter.baulab.info/. arXiv:2510.26784.

Eric Todd, Millicent L. Li, Arnab Sen Sharma, Aaron Mueller, Byron C. Wallace, and David Bau. Function vectors in large language models. In *International Conference on Learning Representations (ICLR)*, 2024.

Jesse Vig, Sebastian Gehrmann, Yonatan Belinkov, Sharon Qian, Daniel Nevo, Yaron Singer, and Stuart Shieber. Investigating gender bias in language models using causal mediation analysis. In H. Larochelle, M. Ranzato, R. Hadsell, M.F. Balcan, and H. Lin (eds.), *Advances in Neural Information Processing Systems*, volume 33, pp. 12388–12401. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper_files/paper/2020/file/92650b2e92217715fe312e6fa7b90d82-Paper.pdf.

A CORPUS

```
<system>
You are a helpful assistant proficient in programming. Always wrap your code in
<code></code> tags. Just the code, no need to comment or explain.
</system>
<user>
Write a python program that implements a function `func` which, given a list of
strings `words`, computes in sequential order:
(1) the total number of characters across all words
(2) the total number of uppercase letters across all words
(3) whether any word is a palindrome
(4) list of words that start with a vowel
</user>
<assistant>
<code>
def func(words: list[str]):
    a = sum(len(w) for w in words)
    b = sum(1 for c in w if c.isupper() for w in words)\n
    c =
```

Figure 5: Illustrative example of the prompt. N subgoals are specified in natural language. For a target frontier, the LM's generated code is prefilled up to a pivot point. On the same figure, we identify the pivot points probed using GOALSCOPE in Appendices C and D with background highlights ("eq" , "var" , "indent" , "prev_end"). Generic <system>, <user>, and <assistant> tags stand in for the LM tokenizer's reserved chat tokens.

We use a fixed corpus of 253 Python specifications to compare GoalScope readouts across head budgets and capture positions (Appendix D), native ablations (Appendices F and G), and model families (Appendix K). To generate the corpus, we sample $N \in \{2, 3, 4\}$ sub-tasks from one of four datatypes: `int`, `str`, `list[int]`, and `list[str]`. Table 2 gives the datatype counts and examples of the full descriptions used in the prompts. Each specification prefills at least one completed sub-task before the frontier g_k, with fixed function and intermediate-variable naming conventions. Figure 5 shows a concrete example, with the four pre-implementation pivots highlighted where we probe for the frontier g_k in most of our evaluations.

Datatype	# Specs	Example sub-task description	Python stub
`int`	65	*whether the number is even*	`a = (n % 2 == 0)`
		the sum of the digits	`a = sum(int(d) for d in str(abs(n)))`
		whether the number is a perfect square	`a = n >= 0 and int(n**0.5)**2 == n`
`list[int]`	60	*the difference between the max and min values*	`a = max(values) - min(values)`
		the integer mean of the values	`a = sum(values) // len(values)`
		the number of distinct values	`a = len(set(values))`
`str`	64	*whether the string is a palindrome*	`a = (s == s[::-1])`
		whether the string has any digits	`a = any(c.isdigit() for c in s)`
		the number of whitespace-separated words	`a = len(s.split())`
`list[str]`	64	*count the number of words longer than 5 characters*	`a = sum(1 for w in words if len(w) > 5)`
		the number of words that start with a vowel	`a = sum(1 for w in words if w and w[0].lower() in "aeiou")`
		the total number of uppercase letters across all words	`a = sum(1 for w in words for c in w if c.isupper())`
Total	253		

Table 2: The locked evaluation corpus and illustrative sub-task descriptions, with example Python stubs from the task bank. Each specification contains two to four sub-tasks; the examples are individual sub-tasks, not complete specifications. We curate both Python and C++ implementation stubs for each sub-task, which we use to prefill the code to the frontier in the language specified by the prompt. In the example stubs, `a` denotes the result variable; `n`, `values`, `s`, and `words` denote the inputs.

B LOCALIZING GOAL HEADS

We locate the *goal heads* with a simpler contrastive setup where we ask the LM to implement two different tasks/predicates g_{src} and g_{dst} in two different programming languages lang_{src} and lang_{dst}. We aim to locate the representations that mediate the task information. Attention heads are a natural candidate as they are the only way a transformer LM can pool information from the context to the current token position[1]. Concretely, we want to locate attention head outputs in the source run $M(p_{\text{src}})$ that, when patched into the destination run $M(p_{\text{dst}})$, will trigger the execution of the source task g_{src} in the destination language lang_{dst}. We use different programming languages ($\text{lang}_{\text{src}} \neq \text{lang}_{\text{dst}}$) to separate the encoding of g_{src} from the surface-level tokens that implement it. Figure 6 shows a pair of example contrastive prompts.

In both source and destination runs, the LM will need multiple tokens to implement g_{src} and g_{dst}. To operationalize the idea in a single forward pass, we use a third prompt that asks the LM to implement the source task g_{src} in the destination language lang_{dst}. In the destination run and the target run, we prefill the implementation with a common prefix such that a successful implementation of g_{dst} in $M(p_{\text{dst}})$ and g_{src} in $M(p_{\text{targ}})$ will generate different next tokens $\text{tok}_{\text{dst}} \neq \text{tok}_{\text{targ}}$. While pairing up contrastive tasks, we choose them such that a potential implementation of g_{src} and g_{dst} in lang_{dst} may share this common prefill.

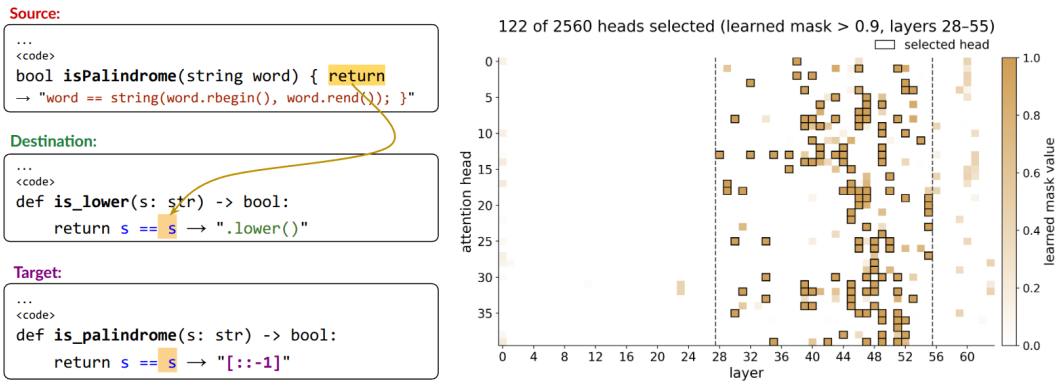


Figure 6: Locating the goal heads. **Left:** source and destination prompts request different predicates in different programming languages. A separate target prompt requests the source predicate in the destination language. The two destination-language implementations share a common prefix and differ in the next token. We patch source head contributions into the destination using the mask-weighted intervention in Equation (2), and jointly train the masks to promote the target continuation. **Right:** the selected 122 heads in Qwen2.5-Coder-32B.

To localize the heads, we learn a joint mask over the heads. $\text{mask}^{\ell j}$ dictates what portion of head $[\ell, j]$'s contribution in the source run we bring over to the patched run. Concretely, we perform the following intervention on the patched run for each attention head:

$$a_{-1}^{\ell j} \leftarrow \text{mask}^{\ell j} a_{\text{src}}^{\ell j} + (1 - \text{mask}^{\ell j}) a_{-1}^{\ell j}. \tag{3}$$

Here, the vector $a_{-1}^{\ell j}$ denotes the OV output of head $[\ell, j]$, at the last token position, and $a_{\text{src}}^{\ell j}$ is the same head's output in the source run, also at the last token position. The terms $\text{mask}^{\ell j}$ are binary values, jointly learned with an objective to match the KL divergence over the next token distributions produced in the target run. Formally, the loss is:

$$\mathcal{L} = \frac{1}{|\mathcal{B}|} \sum_{x \in \mathcal{B}} D_{\text{KL}}\Big(M\left(p_{\text{dest}}\right)\left[\leftarrow a_{\text{src}}\right] \parallel M\left(p_{\text{targ}}\right) \Big) + \lambda_{\text{sparse}} \frac{1}{|\mathcal{H}|} \sum_{[\ell,j] \in \mathcal{H}} \text{mask}^{\ell j}. \tag{4}$$

[1]This assumption is not true for the newer generation of LMs with DeltaNet and our method is not directly applicable there.

We can tune the sparsity loss λ_{sparse} to control the number of heads being selected. $\mathcal{H}$ contains all candidate heads and $\mathcal{B}$ is a training batch. We clip $\text{mask}^{\ell j}$ to stay in the $[0, 1]$ range after each optimization step.

C GoalScope

C.1 Approach

C.1.1 Readout prompts

Figure 7 shows the complete prompts to probe the *faithfulness* and *causality* readouts throughout the paper. Neither probe receives the source prompt being probed as input. The faithfulness prompt begins with an empty assistant message and the causality prompt prefills the assistant response with a fixed C++ function stub.

(a) Faithfulness readout prompt

```
<system>
You are a helpful assistant proficient in programming.
</system>
<user>
What are you doing at this line of code? Keep your answer as brief as possible
</user>
<assistant>
```

(b) Causality readout prompt

```
<system>
You are a helpful assistant proficient in programming.
</system>
<user>
You are implementing a C++ function that checks for a condition. Just write the
function body, without any comments or explanations. Assume all the necessary
headers are already included.
</user>
<assistant>
bool check (auto value){
```

Figure 7: The full prompts used to measure *faithfulness* and *causality* throughout this paper. We start our steering intervention at the last token position and keep performing the same intervention throughout the generation.

C.1.2 Intervention setup

Following the notation used throughout the paper, we call the source context being probed the source prompt p_{src} and the probe prompt p_{probe}. For a selected set of goal heads $\mathcal{G}$, we cache the output of each of these heads at the position t of the token tok_{src} being probed in the source run $M(p_{\text{src}})$. We explore two intervention strategies using these head outputs.

additive_norm. We aggregate the head outputs into a single vector $v_g = \sum_{[\ell,j]\in\mathcal{G}} a_t^{\ell j}$. We use v_g as a steering vector, adding it to a range of layers in the probe run $M(p_{\text{probe}})$. Concretely, we perform the following intervention for each layer ℓ in the selected layer range.

$$h_{-1}^{\ell} \leftarrow h_{-1}^{\ell} + \alpha \|h_{-1}^{\ell}\| \frac{v_g}{\|v_g\|}. \tag{5}$$

The right-hand side denotes the state before this injection; the arrow overwrites it once per layer and generation step.

We perform the same injection repeatedly with greedy decoding. The steering vector direction v_g remains fixed, with the coefficient α rescaling the magnitude at each intervention layer.

replace. This intervention variant instead replaces the goal heads' contributions at every generation step of the probe run $M(p_{\text{probe}})$ with their corresponding contributions from the destination run. Concretely:

$$a_{-1}^{\ell j} \leftarrow \alpha \, a_{\text{src}}^{\ell j} \text{ , for each } [\ell, j] \in \mathcal{G} \tag{6}$$

We use $\alpha = 1.0$ to denote exact replacement, with larger values denoting *boosting* the head contributions to the residual stream at their corresponding places.

We perform the same intervention (in Equation (5) or 6) at each generation step with greedy decoding. The probe prompt does not contain any information about the source context, so the information is communicated solely with the intervention. The main Qwen2.5-Coder-32B model configuration uses the *additive_norm* strategy with $\alpha = .1$, layers 35–45 inclusive. Some of the cross-model comparisons use the *replace* strategy with $\alpha = 1.0$. We did not perform a systematic grid search to select these hyperparameter values.

C.2 Evaluation Setup

Our readout setup generates multiple tokens. We use an LLM judge (`claude-sonnet-4-6`) to check if the readout aligns with the LM's generated code for the frontier g_k. We include the most interesting bits of the faithfulness prompt in Figure 8. The causality prompt is almost identical, just replacing "description" with "C++ code" with some minor adaptations.

```
<system>
…
The question: does the surface anything genuinely relevant to what the CODE is doing — a relevant operation, quantity,
sub-step, or property of the code? Even a correct fragment of the right idea is a SUCCESS, so grade leniently — except that a
read-out which has collapsed into a degenerate repeating loop is not a success.

Decide in this order: check MALFORMED and HEAVY_LOOP first, and only then YES/NO/NO_INFO:
…
  MALFORMED  - the description is contentless: empty, pure punctuation / garbage, or repetition carrying no relevant meaning.
  HEAVY_LOOP - the description is dominated by a short repeating cycle
  YES        - the code contains SOME operation, quantity, or sub-step genuinely relevant to what the code does, however
partial, buggy, or incomplete — and it is not merely a short repeating loop.
  NO         - the code is coherent and about a specific thing, but that thing is UNRELATED to the what the code does. None of
the conceptual sub-parts have been identified correctly.
  NO_INFO    - the description is coherent but too generic to indicate any specific aspect of the code. (e.g. 'filtering from
a list of strings').
…
Put your final answer in <verdict>{ans}<verdict> tags to facilitate automated parsing.
</system>
<user>
Code: {impl_span}
Description: {readout}
</user>
```

Figure 8: Prompt for the LLM judge to evaluate *faithfulness*. Note that we ask the judge to make a subjective decision of whether the readout is a non-trivial sub-part of the LM's generated code in response to g_k.

We count the judge's response of `YES` as success and all other responses, including cut-off/corrupted responses from the judge, as failures. Note that in the oracle prompt we ask the judge to make a subjective judgement of whether the readout captures a non-trivial portion of the upcoming LLM-generated code. So, the readout rates do not reflect 100% alignment with the code (or execution accuracy of the C++ code in causality).

D ARE GOAL HEADS ANYTHING SPECIAL? HOW MANY DO WE NEED FOR A DECENT READOUT?

D.1 COMPARING AGAINST FULL ATTENTION OUTPUTS AND RESIDUAL STATES

Are the selected *goal heads* important for the readout, or could we get a similar description by steering with a larger portion of the LM's state? To check this, we compare their aggregated contributions against the full attention module outputs and the residual stream. We use the same source prompts, readout prompts, and four pre-implementation pivots highlighted in Figure 5.

All three kinds of source use norm-relative additive steering with $\alpha = 0.1$, at probe layers 35–45. Concretely, we sum the 122 goal-head contributions, or average the full attention outputs or residual states over the specified source layers. We then normalize this vector before injecting it, following Equation (5). This keeps the intervention magnitude comparable despite the different numbers of heads and layers contributing to each source.

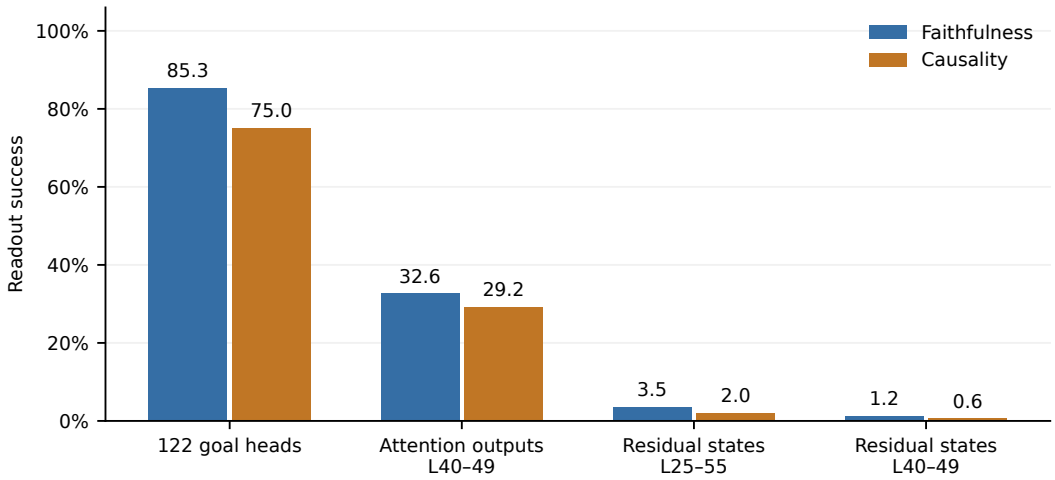


Figure 9: **Comparison of readout sources.** We compare the goal-head contributions with full attention outputs and residual states, using norm-relative additive steering at $\alpha = 0.1$. Each bar averages the four pre-implementation pivots over the same 253 specifications ($n = 1,012$ readouts). The source layer ranges are inclusive. The first implementation token is excluded.

We find that the goal-head contributions give substantially better readouts: 85.3% faithfulness and 75.0% causality, compared with 32.6% and 29.2% from the full attention outputs at layers 40–49. The residual baselines have low success under this intervention. So, selecting the goal heads helps expose the upcoming sub-goal to the readout prompt. The comparison holds the steering setup fixed; we did not separately tune its strength or layer range for each source. We return to extracting goal information from the residual with OVReader in Appendix I.

D.2 READOUTS WITH DIFFERENT NUMBERS OF HEADS

How many of these heads do we need for a decent readout? We vary the selected set from 5 to 300 heads and repeat both readouts at each pivot. The main figure (Figure 2, left) averages over the four pre-implementation pivots. Figure 10 shows the individual positions, so we can distinguish the effect of the head budget from when the goal becomes readable.

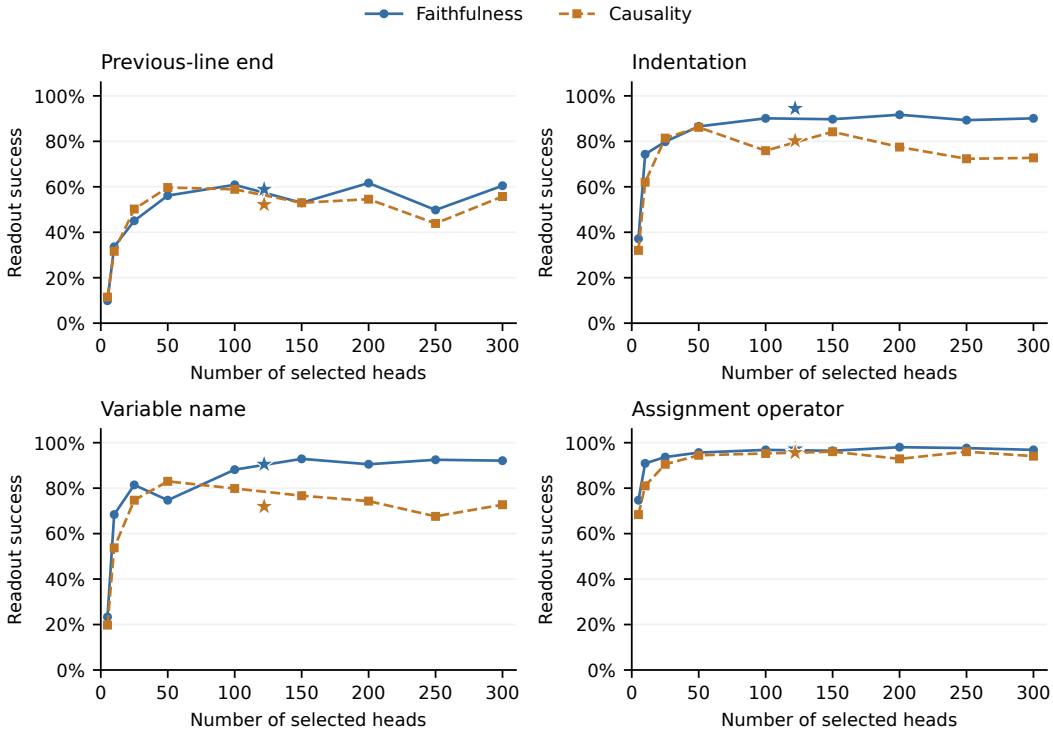


Figure 10: **Head budget at each pre-implementation pivot.** Each point measures readout success on 253 specifications, with norm-relative additive steering at $\alpha = 0.1$. Stars mark the separately selected 122-head set. The four panels contribute equally to the aggregate curve in Figure 2; none includes an implementation token.

At the assignment pivot, 25 heads give 93.7% faithfulness and 90.5% causality. Averaged over all four pivots, the same set gives 75.0% and 74.2%. The 122-head set gives 85.3% and 75.0% on this average. Thus, a small fraction of heads already supports effective readout, although performance depends on the pivot. Adding heads does not improve the readouts monotonically. These selections come from different mask thresholds and snapshots and are not strictly nested, so this sweep does not isolate head count from head membership.

E ACROSS MODEL COMPARISONS

We next check whether the same setup works across LM sizes and training variants. We localize the *goal heads* separately in each model and use that model for both the source and probe runs. All comparisons below use the same 253 specifications and four pre-implementation pivots. We use the *replace* strategy with $\alpha = 1.0$ throughout, so the intervention acts at each selected head's own layer. This differs from the additive setup used in Appendix D.

Model	Goal heads	Faithfulness	Causality
Coder 32B	122	0.789	0.735
Coder 14B	88	0.519	0.305
Instruct 32B	108	0.238	0.441
OLMo +RL	93	0.435	0.622
OLMo SFT	91	0.529	0.452

Table 3: Readout success across models, averaged over the four pre-implementation pivots ($n = 1,012$ per model and mode). Each model uses its own selected heads. The five-position results, including the first implementation token, and exact selection names are in Appendix K.

E.1 MODEL SIZE

We compare Qwen2.5-Coder-Instruct at 14B and 32B parameters. The 32B model has better readouts on both measures: faithfulness increases from 51.9% to 78.9%, and causality from 30.5% to 73.5%, averaged over the four pivots. Figure 11 shows that the larger model's advantage is already visible before the assignment operator. We use 88 selected heads in the 14B model and 122 in the 32B model; this is a comparison of the localized readouts in each model, rather than a matched-head-budget test of size alone.

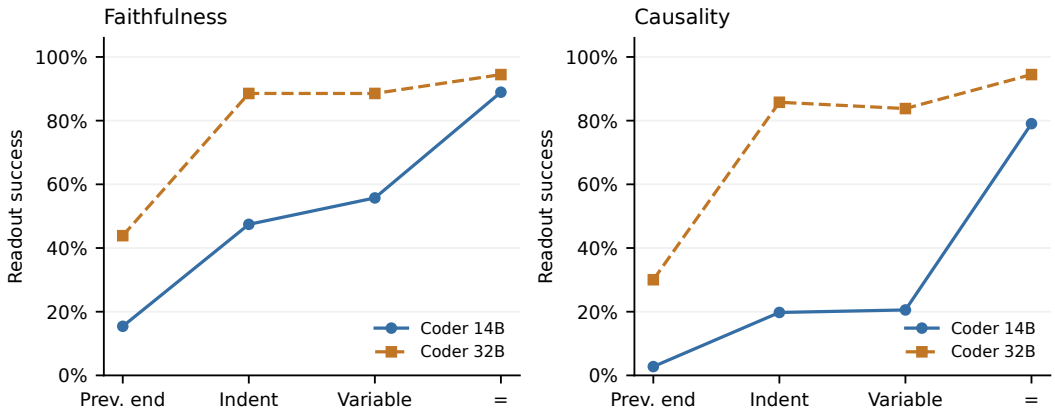


Figure 11: **Qwen2.5-Coder-Instruct: 14B vs. 32B.** Both models use head replacement at $\alpha = 1.0$. Each point contains 253 readouts. The 32B model gives better readouts at all four pivots in both modes.

E.2 CODE-FINETUNING HELPS

We compare Qwen2.5-Coder-32B-Instruct with Qwen2.5-32B-Instruct to check how the readouts differ between code-specialized and general instruction models of the same size. The Coder model gives better readouts at every pivot in Figure 12. Averaged over the four pivots, faithfulness increases from 23.8% to 78.9%, and causality from 44.1% to 73.5%. The comparison supports the use of a code-specialized model for these code tasks. It does not isolate code finetuning from other differences between the checkpoints or the independently selected head sets (108 vs. 122 heads).

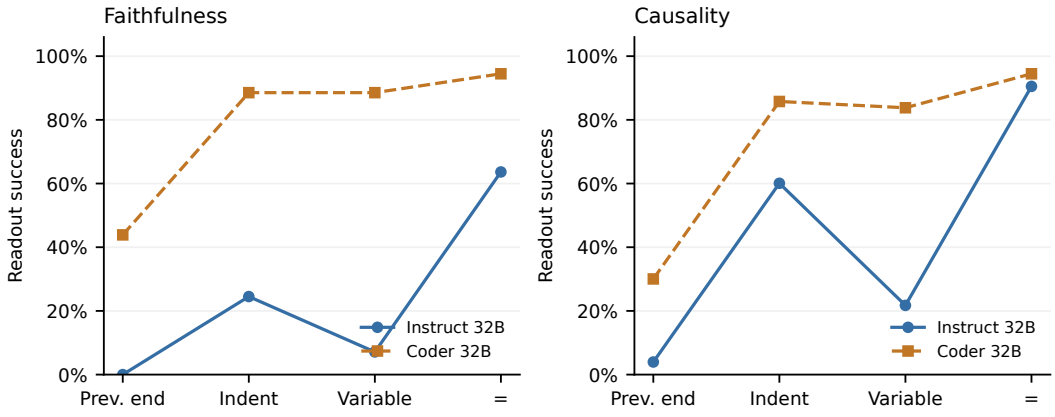


Figure 12: **Qwen2.5-Instruct: Coder vs. non-Coder at 32B.** The same prompts and replacement strategy are used in both models, with model-specific head sets. The advantage of the Coder model is visible in faithfulness and causality.

E.3 RL-FINETUNING ALSO HELPS

We compare OLMo-3.1-32B-Instruct with its SFT checkpoint, using 93 and 91 selected heads, respectively. The two readout measures move in different directions: causality increases from 45.2% to 62.2%, while faithfulness decreases from 52.9% to 43.5%. Thus, the improvement here is in triggering an implementation of the source sub-goal; the instruction model does not verbalize it more faithfully on average.

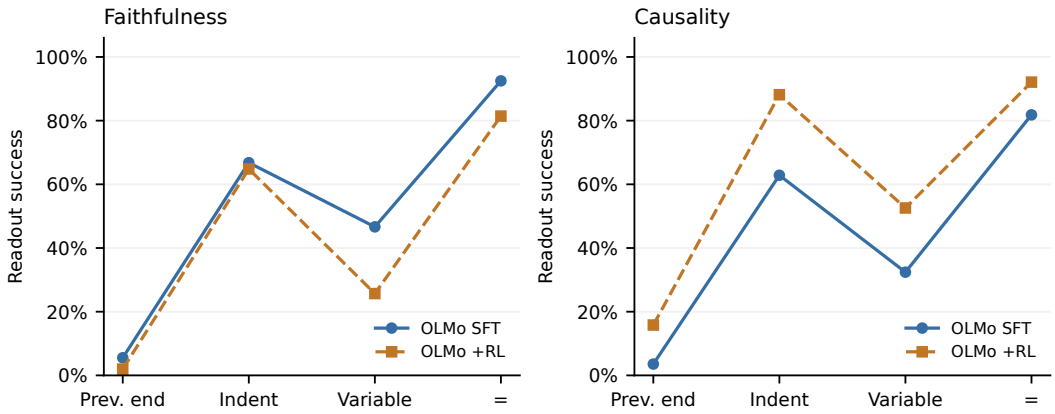


Figure 13: **OLMo-3.1-32B: instruction vs. SFT checkpoint.** Head replacement uses $\alpha = 1.0$ in both models. Causality is higher for the instruction model at every pivot, whereas faithfulness is lower.

The instruction model often responds that it cannot answer without seeing the code. As a simple check on this wording, 402 of its 1,012 pre-implementation faithfulness readouts contain "I cannot" or "I can't", compared with 11 for SFT. The rate depends on the pivot: 192 of 253 readouts at the previous-line end contain one of these phrases, compared with 40 of 253 at the assignment operator. These are literal phrase counts, not an oracle classification of refusals.

Table 4 shows four matched examples at the assignment pivot. Some responses decline to answer but still mention the source operation, which can receive a `YES` under the partial-content rubric in Appendix C. Other responses provide no task information. This behavior can affect faithfulness without implying that the same state cannot trigger the source operation in the causality probe. The examples illustrate the difference in wording; they do not establish that it explains the entire faithfulness gap.

Model	Readout	Verdict
whether the number is even (`int_python_003`)		
SFT	I'm checking if a number is even.	`YES`
Instruct	I cannot answer without knowing the specific line of code.	`NO_INFO`
whether the number is even (`int_python_001`)		
SFT	I'm checking if a number is even.	`YES`
Instruct	I can't determine if a number is even or odd from this line alone; I need the full context.	`YES`
the sum of all values (`list[int]_python_007`)		
SFT	I am summing all the elements in the list.	`YES`
Instruct	I cannot access or process the specific line of code or data you are referring to without additional context. Please provide the code or clarify the question.	`NO_INFO`
whether any word is a palindrome (`list[str]_python_000`)		
SFT	I am checking if the variable `is_valid` is `False`.	`NO`
Instruct	I'm checking if any element in a list meets a certain condition (e.g., if any string is a palindrome).	`YES`

Table 4: **Matched OLMo faithfulness readouts at the assignment pivot.** Four selected examples with their saved oracle verdicts. The complete readouts are shown, including refusal-like wording and a case where the instruction model succeeds while SFT does not. These examples are illustrative; the rates in Table 3 use all specifications.

F SUFFICIENCY OF GOAL-HEAD ACCESS

We test whether the model can implement the frontier when only the selected heads retain access to the frontier-description and progress span. At the final prompt token and subsequent decoding positions, heads outside the spared set in layers 20–63 cannot attend to this span. The blocked key interval starts one token before the frontier description and ends at the prompt's final assignment token. Earlier query positions and layers below 20 remain unchanged.

```
<system>
You are a helpful assistant proficient in programming. Always wrap
your code in <code></code> tags. Just the code, no need to comment
or explain.
</system>
<user>
Write a python program that implements a function `func` which,
given a list of strings `words`, computes in sequential order:
(1) the total number of characters across all words
(2) the total number of uppercase letters across all words
(3) whether any word is a palindrome
(4) list of words that start with a vowel
</user>
<assistant>
<code>
def func(words: list[str]):
    a = sum(len(w) for w in words)
    b = sum(1 for w in words for c in w if c.isupper())
    c =
```

Figure 14: **Sufficiency intervention.** Only spared heads retain access to the green span in layers 20–63. The purple box marks the final `=` query; the same restriction continues during generation. This illustrative prompt targets a palindrome test.

Condition	122-head set	250-head set
Clean	0.925	0.925
Spare goal-head access	0.806	0.913
Spare random-head access	0.091	0.095

Table 5: Frontier implementation success on 253 specifications under span restriction. Goal and random rows differ in the set whose access is spared.

For the 122-head comparison, each condition spares 122 of the 1,760 heads in layers 20–63. For the broader selection, the goal condition spares 236 in-band heads and the random condition spares 170; selected heads below layer 20 lie outside the restricted band. Thus the broader control does not match the number of spared in-band heads. Random sets are sampled from the corresponding depth pool, without matching counts at each layer. The intervention tests access to this span, rather than isolation of goal heads from all other network computations.

Condition	Continuation after `b =`	Verdict
Clean	`(n % 2 == 0)`	YES
Goal heads	`(n % 2 == 0)`	YES
Random heads	`(n * (n + 1)) // 2`	NO

Table 6: **An observed sufficiency example.** Exact continuations for `int_python_000`: digit count has been completed and the frontier is evenness. The 250-head goal condition preserves the clean implementation; the random condition implements a different computation. This example is separate from the illustrative palindrome prompt above.

G NECESSITY OF GOAL-HEAD ACCESS

We restrict selected heads' access at the final prompt token and all subsequent decoding positions. The sink intervention allows attention only to key position zero, denoted by generic <bos> in Figure 15 (<|im_start|> in Qwen). It preserves the sink value contribution and is not equivalent to zeroing a head. Two further interventions block the frontier description (including its preceding token), or extend that span through the prompt's final assignment token. Earlier query positions remain unmodified.

```
<bos><system>
You are a helpful assistant proficient in programming. Always wrap
your code in <code></code> tags. Just the code, no need to comment
or explain.
</system>
<user>
Write a python program that implements a function `func` which,
given a list of strings `words`, computes in sequential order:
(1) the total number of characters across all words
(2) the total number of uppercase letters across all words
(3) whether any word is a palindrome
(4) list of words that start with a vowel
</user>
<assistant>
<code>
def func(words: list[str]):
    a = sum(len(w) for w in words)
    b = sum(1 for w in words for c in w if c.isupper())
    c =
```

Figure 15: **Necessity intervention with sink-only attention.** Selected heads cannot attend to the orange span and attend only to position zero. Purple marks the final assignment query. The same sink restriction continues throughout decoding.

	122-head set		250-head set	
Blocked keys	Goal	Random	Goal	Random
Frontier description	0.881	0.925	0.806	0.925
Frontier description through pivot	0.372	0.925	0.249	0.925
All except position zero	0.854	0.925	0.158	0.921

Table 7: Frontier implementation success under three access restrictions ($n = 253$). The clean baseline is .925. Each goal condition has a random-head control with the same total selected count, sampled from the corresponding depth pool.

Blocking the frontier description alone has a smaller effect than blocking the description and progress span together. The sink intervention substantially reduces success for the broader 250-head set, whereas the 122-head core produces a smaller reduction. These outcomes depend on head membership, depth coverage, and the access restriction being tested.

Condition	Continuation after `b =`	Verdict
Clean	`(n % 2 == 0)`	YES
Goal heads, task only	`n % 2 == 0`	YES
Goal heads, task + progress	`sum(1 for _ in str(n) if _.isdigit())`	NO
Goal heads, sink	`nn``	MALFORMED
Random heads, sink	`(n % 2 == 0)`	YES

Table 8: **An observed necessity example.** The same evenness frontier as Table 6, using the 250-head set. Task-only restriction preserves evenness, task-and-progress restriction repeats digit count, and sink restriction produces a malformed continuation. The random sink control preserves the correct operation.

H TASK AND QUERY TRANSFER

Task-span KV transfer uses per-query-head output routing to avoid transferring to every query head sharing a KV head. Task spans are padded to equal length and source keys are re-rotated to destination positions; equal lengths do not imply identical absolute starts. The KV patch is active across prefill positions and decoding. Query transfer patches the specified pivot tokens and reuses the source pivot query during the steering window (32 forward passes in the evaluation CLI, including prefill). The reported transfer configurations also amplify selected OV contributions with coefficient 2; this differs from the GoalScope probe's coefficient 0.1.

The principal multi-predicate 250-head runs report approximately 94% source-directed generation for query transfer ($n = 2048$) and 99% for KV transfer at the assignment pivot ($n = 512$), spanning four datatypes and four language directions. The latter uses no non-goal-head ablation. These rates count source-or-both oracle labels and do not imply perfect program correctness. GoalScope's patched source readout is a separate outcome. Q-filter examples exhibit readout/behavior disagreement; aggregate readout success should not be treated as a substitute for implementation success.

The qualitative task-order example in Figure 19 uses unequal source/destination progress. It should not be described as an identical-progress experiment. Broader conclusions about query content require a systematic crossed design. We make no claim that these interventions demonstrate a particular set-subtraction operation.

H.1 KV TRANSFER: AGGREGATE AND DATATYPE RESULTS

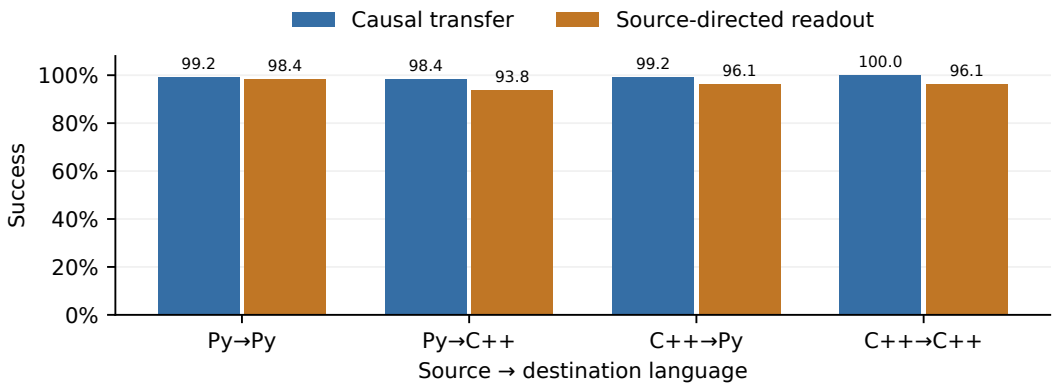


Figure 16: **KV transfer across languages.** Each language direction pools 128 pairs across four datatypes. Causal transfer counts source-or-both implementation verdicts; source-directed readout counts exclusively source-labeled patched descriptions. Both use all pairs in the denominator. Runs use 250 heads, assignment pivots, task-span padding, OV amplification 2, and no non-goal-head ablation.

The source-directed readout score counts patched descriptions judged exclusively source-directed, with all examples in the denominator. The stored `valid` flag asks whether the original source and destination readouts match their respective frontiers, allowing a both-goal verdict; its rates are 99.2%, 96.9%, 96.1%, and 94.5% for Python–Python, Python–C++, C++–Python, and C++–C++, respectively. The stored `shifted` flag counts patched source-only or both-goal readouts, without requiring either clean readout to be correct. Its unconditional rates happen to match the source-directed readout scores here because none of these patched readouts is labeled both. The evaluation script also reports shift conditional on `valid`; that denominator differs from the unconditional scores shown here.

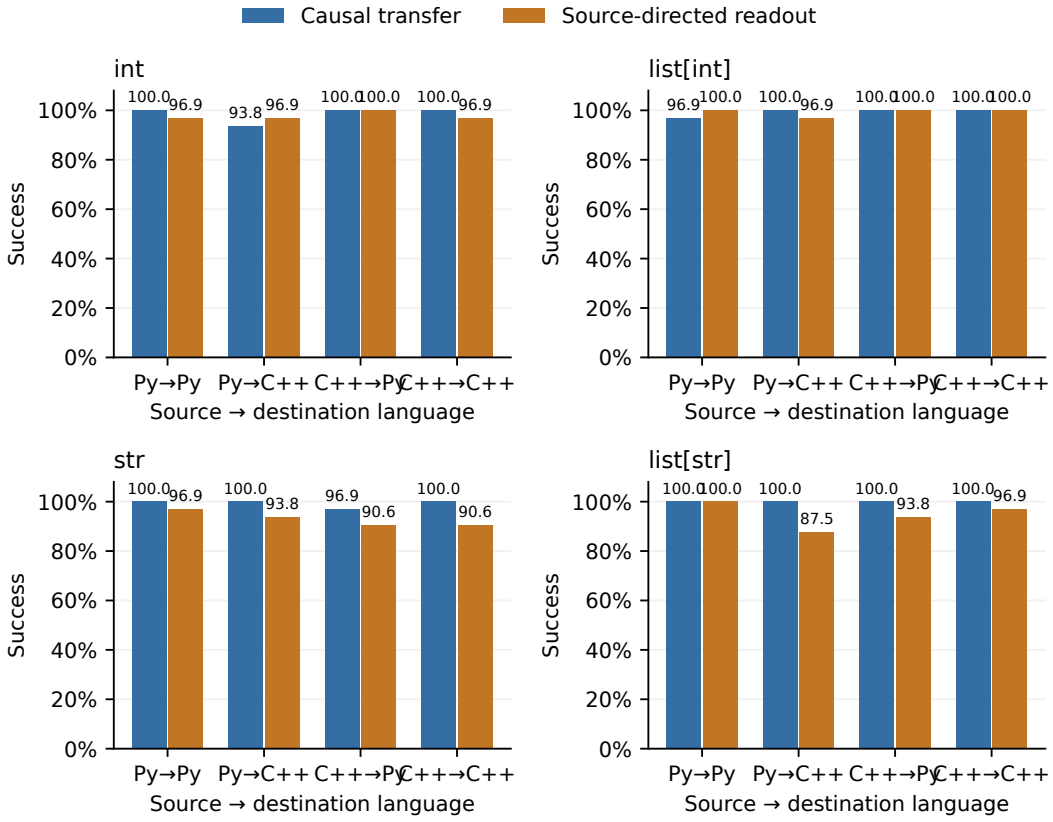


Figure 17: **KV transfer by datatype.** Breakdown of Figure 16, with 32 pairs per language direction in each panel. Metrics, settings, and denominators are unchanged.

H.2 A FREE-FORM KV-TRANSFER EXAMPLE

(a) Source run, $M(p_{\text{src}})$

```
...
returns true if the shortest word starts with a vowel
...
<code>
auto func(std::vector<std::string> words){

    std::vector<std::string> sorted_words = words;
    std::stable_sort(sorted_words.begin(),
sorted_words.end(), [](const std::string& a, const
std::string& b){ return a.size() < b.size(); });
    char first_char = std::tolower(sorted_words[0][0]);
    return first_char == 'a' || first_char == 'e' ||
first_char == 'i' || first_char == 'o' || first_char == 'u';
}
```

↳ **readout** ' ' => *I'm checking if the string starts with a vowel.*

(b) Destination run, $M(p_{\text{dst}})$

```
...
returns true if the longest word is a palindrome
...
<code>
def func(words: list[str]):
    res = sorted(words, key=len)
    return res[-1] == res[-1][::-1]
</code>
```

↳ **readout @** ' return' => *I'm checking if the string is equal to its reverse.*

(c) Patched run, $M(p_{\text{dst}} \mid \leftarrow \{K, V\}_{\text{src}})$

```
...
    return res[0][0].lower() in 'aeiou'
</code>
```

↳ **readout @** ' return' => *I'm checking if the first character of the string is in 'aeiouAEIOU'.*

Figure 18: **KV transfer after a shared initial sub-task.** The source and destination both sort the words by length (blue), then pursue different tasks: a vowel test on the shortest word versus a palindrome test on the longest word. Replacing the destination task-span keys and values available to the goal heads with those from the source changes the continuation to a vowel test on the shortest word. The readout shifts accordingly. Source, native destination, and patched continuations are red, green, and purple; yellow marks their pivots. This is an illustrative slide example, separate from the scored multi-predicate cohort. In the evaluated implementation, the fixed source span is routed to selected query heads at every forward pass; other heads retain destination KV access.

H.3 QUERY TRANSFER UNDER A TASK-ORDER CONTRAST

Let A count characters, B count uppercase letters, C test whether any word is a palindrome, and D count vowel-initial words. The source orders (A, B, C, D) and has completed (A, B); the destination orders (A, B, D, C) and has completed only A. Its native next task is therefore B. A transferred state encoding only completed tasks (A, B) would make D next under destination order, whereas the source's next goal is C. The slide example shows a query-patched continuation implementing C, accompanied by a palindrome readout. This is a qualitative contrast, not a factorial evaluation of progress and goal identity.

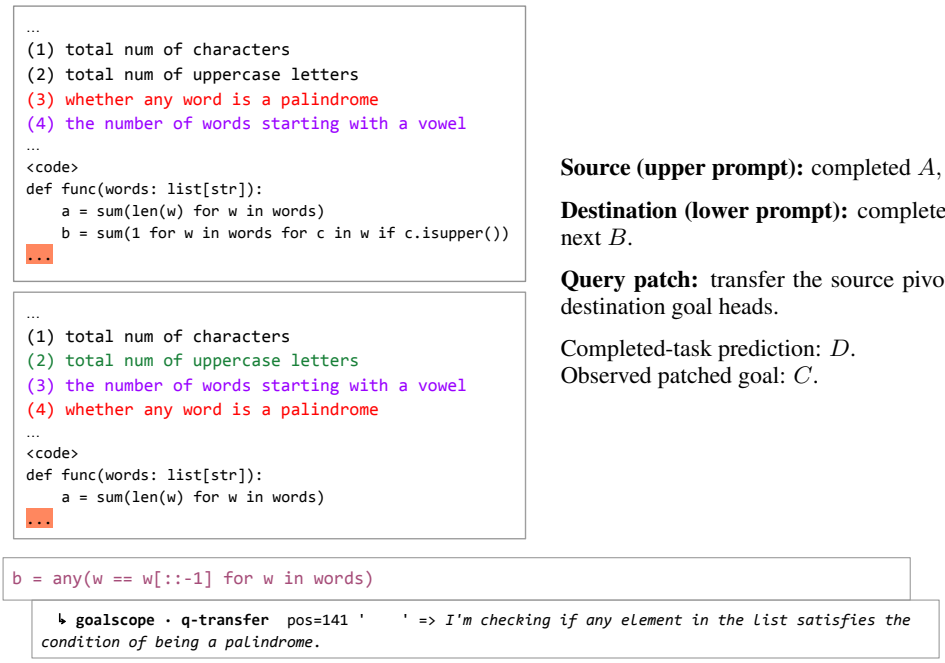


Figure 19: **A query-transfer contrast from the exploratory slides.** The source and destination contain the same four tasks in different orders and have unequal prefills. The patched destination computes whether any word is a palindrome, matching the source's upcoming goal. Ellipses in the prompt panels mark the capture/patch boundary; the saved readout identifies position 141. This example does not establish that query states contain only goal information.

I OVREADER

I.1 CONSTRUCTING THE READOUT DIRECTION

Let $h_{L,t} \in \mathbb{R}^d$ be the residual state after source layer L at position t. For selected head j, let $O_j \in \mathbb{R}^{d \times d_h}$ be its block of the attention output projection, and let $V_j \in \mathbb{R}^{d_h \times d}$ be its value projection. Query heads that share a KV head use the corresponding shared value map. Incorporating the head's pre-attention RMSNorm gain γ_j, define

$$B_j = V_j \operatorname{diag}(\gamma_j), \qquad A_j = O_j B_j. \tag{7}$$

The normalization's scalar denominator is immaterial after normalizing each transformed vector. OVReader applies these weight maps directly to the residual at the chosen source layer, without computing a new attention distribution:

$$u_j(h) = \frac{A_j h}{\max(\|A_j h\|_2, \epsilon)}, \qquad g_{\mathrm{OV}}(h) = \frac{1}{|H|} \sum_{j \in H} u_j(h), \qquad \epsilon = 10^{-8}. \tag{8}$$

This is a weights-only readout, not the head's actual attention-weighted output. Per-head normalization also makes it different from a single linear map obtained by averaging the A_j. We inject g_{OV} with norm-relative coefficient .15 into probe layers 35–45 throughout decoding. The actual head-output reference uses .1.

I.2 FAITHFULNESS AND CAUSALITY ACROSS LAYERS

Figures 20 and 21 give separate sweeps for both modes at all five capture positions. Each faint curve corresponds to one datatype; the solid black curve averages the four datatype rates. The dashed reference uses actual goal-head outputs at the same pivot.

These plots use the complete archived evaluation: 64 specifications per datatype, 19 source layers, and a head-output reference. The same 256 specifications occur at every layer and pivot in both modes. This is the cohort used in the main OVReader figure, with rule-based implementation spans. The newer locked-corpus rerun is incomplete for causality, so we do not mix its records into these comparisons. The faint curves show differences across datatypes, not uncertainty over independent runs. Causality records include trivial-output sanitization.

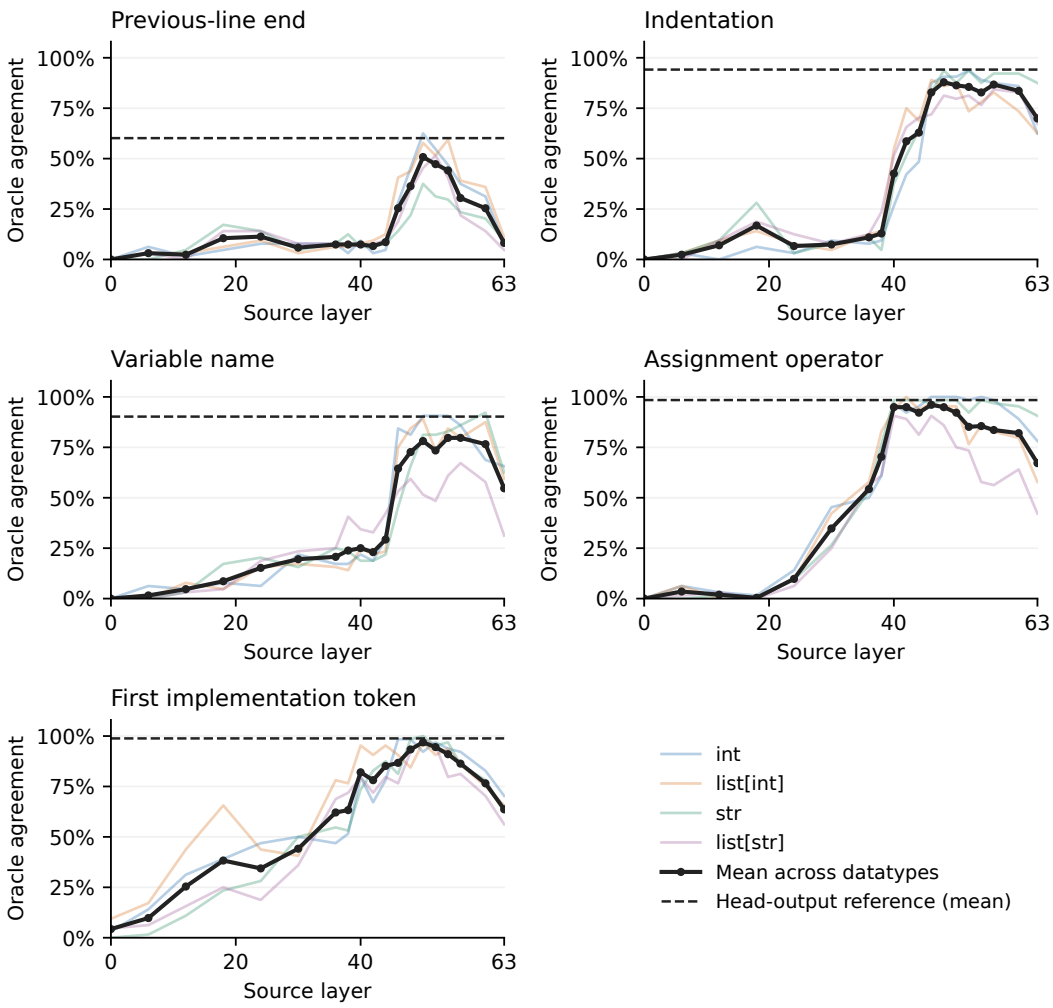


Figure 20: **OVReader faithfulness at all five capture positions.** Each datatype contributes 64 specifications per point; the mean pools all 256. Source-layer indices are zero-based. The dashed line is mean faithfulness from the actual head contributions at the same position. The first implementation token is included as a comparison after implementation begins.

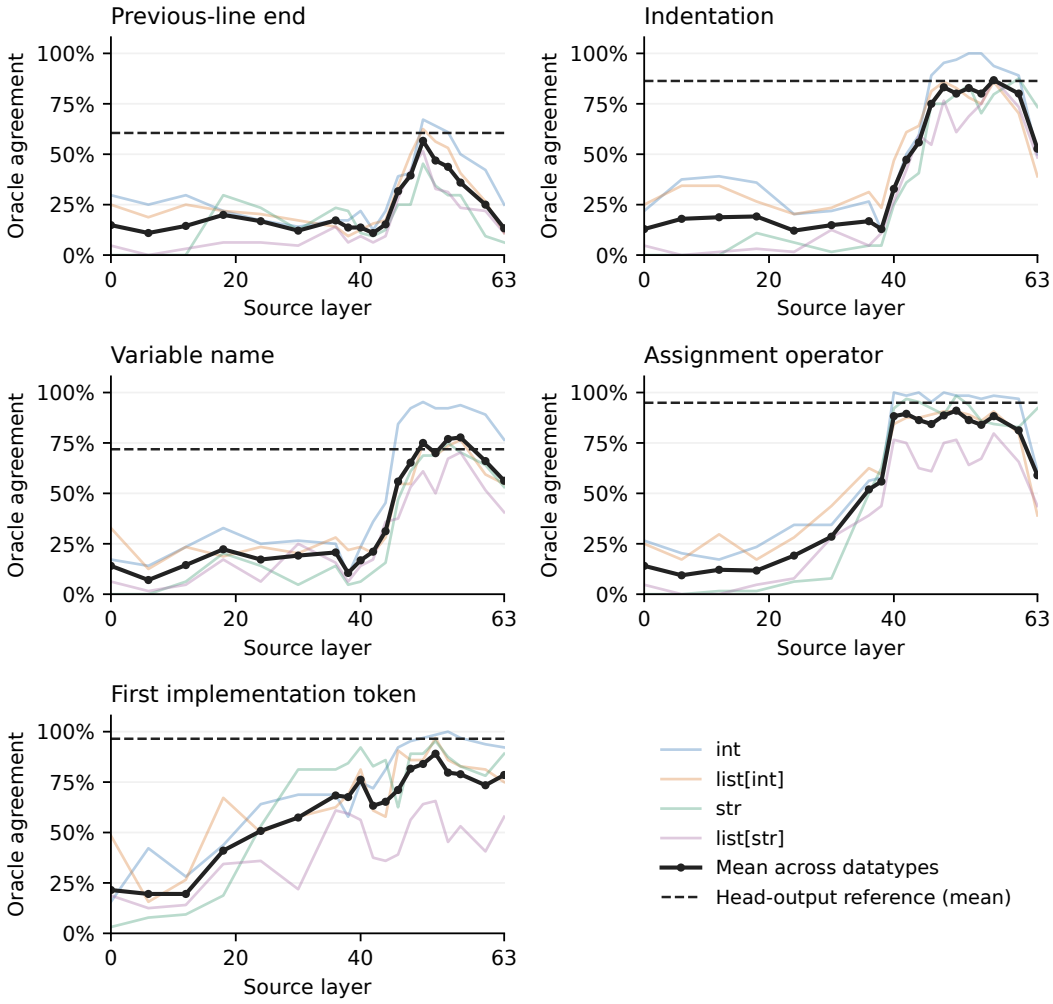


Figure 21: **OVReader causality at all five capture positions.** The cohort, source layers, and aggregation match Figure 20. Scores measure oracle agreement of the C++ probe implementation with source code, rather than execution-based accuracy. The dashed line uses actual head contributions.

I.3 ALTERNATIVE TRANSFORMATIONS OF THE RESIDUAL

The exploratory notebook compares the OV map with projections onto read and write subspaces and an eigenvector-based projection. With $^+$ denoting the Moore–Penrose pseudoinverse, the per-head transformations are:

Variant	Transformation of h	Construction
OV	$O_j B_j h$	Apply the value and output maps.
Write projection	$O_j O_j^+ h$	Project onto the output map's column space.
Read projection	$B_j^+ B_j h$	Project onto the value map's row space.
Eigen projection	$Q_j Q_j^\top h$	Project onto an orthonormal basis built from leading eigenvectors.

Table 9: Residual transformations compared in the saved exploratory example. Each variant normalizes per-head vectors and averages them before the same norm-relative injection.

The write and read spaces equal the column and row spaces of A_j when B_j has full row rank and O_j has full column rank, respectively. For the eigen variant, the notebook diagonalizes $B_j O_j$, selects 64 eigenvectors by eigenvalue magnitude, maps them into residual space using O_j, and constructs Q_j by QR decomposition of their real and imaginary parts. This QR step is not rank-truncated: redundant or zero columns can introduce additional basis directions. Thus the saved exploratory variant need not be a projection strictly onto the leading eigenspace.

```
(1) whether any word is empty
(2) the number of words longer than five
characters
(3) the number of words that start with a vowel

<code>
def func(words: list):
    a = any(len(w) == 0 for w in words)
    b = sum(1 for w in words if len(w) > 5)
    c = sum(1 for w in words if
w.lower().startswith(('a', 'e', 'i', 'o', 'u')))
    return a, b, c
</code>
```

Figure 22: **Source prompt for the transformation comparison.** The source has counted empty and long words; at the highlighted assignment it is about to count vowel-initial words. Only the prompt panel is reproduced here; the table below gives the full saved layer-40 readouts.

Variant	Readout	Verdict
Actual heads	Checking if the character is a vowel.	YES
OV	I'm checking if the character is a vowel.	YES
Write projection	I'm checking if a specific condition is met in the code, typically involving a variable or a function that checks for a particular value or pattern.	NO_INFO
Read projection	I'm checking if the current character is a vowel (assuming the line is `sum(1 for char in line if char.startswith(tuple([c for c in line if c.lower().startswith('aeiou')]))`.	YES
Eigen projection	This line checks whether the variable is a boolean (checking if it is a boolean or not) ```python isinstance(x ``	NO

Table 10: **One saved comparison at layer 40 and the assignment pivot.** Full readouts from `07.1_ov_probe_comparison.ipynb`. Residual variants use $\alpha = .15$; the actual-head reference uses .1. Under the partial-content rubric, YES need not recover the counting operation. This example is not an aggregate comparison of the variants.

J Attention to Task Descriptions

The query-transfer example in Figure 19 also provides a qualitative view of attention to the destination task list. Its native frontier is the uppercase-count task; after transferring source queries, the continuation and readout instead pursue the palindrome predicate. The slide highlights both descriptions in the task list. This visualization illustrates the association between the heads' access to task tokens and the operation expressed by the continuation.

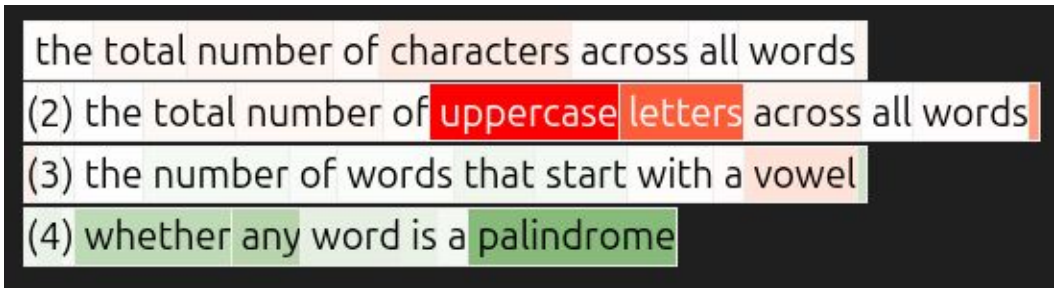


Figure 23: Task-description highlights accompanying the query-order experiment. The destination's native uppercase-count frontier is highlighted in red and the transferred palindrome goal in green. The slide supplies no calibrated color scale; this panel should be read as a qualitative visualization rather than a quantitative attention-mass comparison.

A quantitative account requires a specified attention statistic, token-span definition, and aggregation across heads and examples. We leave comparison with independently localized gaze or concept-copying heads to a separate evaluation.

K Generalization Across Model Families and Sizes

We repeat localization and evaluation within five models: Qwen2.5-Coder at 14B and 32B parameters, Qwen2.5-32B-Instruct, and OLMo-3.1-32B instruction and SFT variants. Each model uses its own selected heads and its own probe generation. This tests replication of readable and transferable task information, not activation transplantation between models.

K.1 Matched evaluation and head selection

All ten runs use the same 253 specification identifiers and five capture positions, yielding 1,265 records per model and mode. Rates below are recomputed from saved records. The intervention is head replacement with source coefficient 1 and steering throughout decoding; the probe generation budget is 64 tokens. Replacement operates at the selected heads' own layers and ignores the additive steer-band setting. Thus the results are not directly interchangeable with the main norm-relative additive scores.

Model	Heads	Selected layer range	Faithfulness	Transfer
Qwen Coder 32B	122	28–55	0.823	0.783
Qwen Coder 14B	88	18–47	0.598	0.401
Qwen Instruct 32B	108	29–55	0.308	0.540
OLMo Instruct 32B (+RL)	93	11–35	0.495	0.689
OLMo Instruct 32B (SFT)	91	11–34	0.595	0.533

Table 11: Within-model replication on the locked corpus, pooled over capture positions. Head sets are model-specific. A selected layer range gives its endpoints, not a claim that every layer is represented.

The Qwen sets contain 122, 88, and 108 heads, respectively. The OLMo comparison uses 93 instruction-model heads and 91 SFT-model heads from the candidate band 10–36, using the exported threshold selections rather than the older 130/79-head pair. Their counts are approximately matched, not identical. Head membership and its layer distribution remain model-specific. Full model identifiers and selection names are listed below for reproducibility.

`Qwen/Qwen2.5-Coder-32B-Instruct`
Selection: `goal`.

`Qwen/Qwen2.5-Coder-14B-Instruct`
Selection: `goal`.

`Qwen/Qwen2.5-32B-Instruct`
Selection: `goal`.

`allenai/Olmo-3.1-32B-Instruct`
Selection: `json:goal__lam20_thr1.0_band10-36`.

`allenai/Olmo-3.1-32B-Instruct-SFT`
Selection: `json:goal__lam20_thr0.6_band10-36`.

K.2 Results by capture position

Model	Metric	Prev. end	Indent	Variable	Assignment	First impl.
Qwen Coder 32B	Faith.	0.439	0.885	0.885	0.945	0.960
Qwen Coder 32B	Transfer	0.300	0.858	0.838	0.945	0.972
Qwen Coder 14B	Faith.	0.154	0.474	0.557	0.889	0.917
Qwen Coder 14B	Transfer	0.028	0.198	0.206	0.791	0.783
Qwen Instruct 32B	Faith.	0.000	0.245	0.071	0.636	0.589
Qwen Instruct 32B	Transfer	0.040	0.601	0.217	0.905	0.937
OLMo Instruct 32B (+RL)	Faith.	0.020	0.648	0.257	0.814	0.735
OLMo Instruct 32B (+RL)	Transfer	0.158	0.881	0.526	0.921	0.957
OLMo Instruct 32B (SFT)	Faith.	0.055	0.668	0.466	0.925	0.862
OLMo Instruct 32B (SFT)	Transfer	0.036	0.628	0.324	0.818	0.858

Table 12: Exact per-position rates for the model comparison in Figure 24.

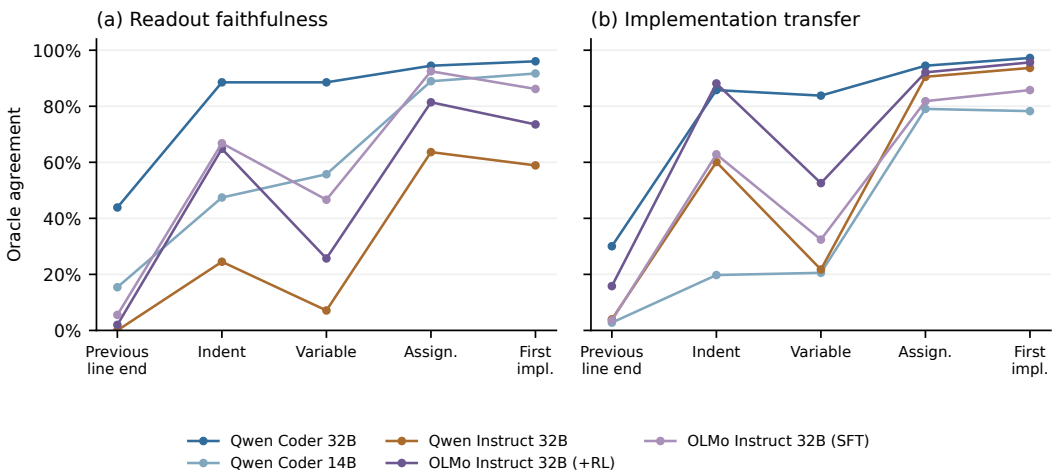


Figure 24: Readout faithfulness and implementation transfer within each model across capture positions (253 records per point). The first implementation token is separated from the four earlier positions. The common scale exposes differences in timing and metric, which pooled rates can obscure.

K.3 Interpretation

The Coder-32B configuration scores above Coder-14B on both pooled metrics; it also outperforms the generic Qwen-32B instruction configuration on these code tasks. For OLMo, the instruction model has higher implementation-transfer scores, while the SFT model has higher readout faithfulness. Replication therefore does not imply identical performance or a single ordering of models across metrics.

These comparisons vary checkpoints, head sets, and potentially source-generation competence together. They support the presence of the phenomenon across families and sizes, but do not isolate causal effects of scale, code training, or RL. Capturing at equivalent named boundaries also need not mean identical tokenization across models. The approximate OLMo budget matching reduces one disparity without controlling every difference. All scores use the partial-content oracle rubric in **??**, rather than execution-based program correctness.

L READOUTS WITHOUT ENUMERATED SUB-TASKS

The controlled evaluation names the component predicates in the instruction. The examples below instead read intermediate operations from familiar programs whose component steps are not separately enumerated. They are qualitative illustrations from the exploratory slides, not part of the 253-specification evaluation.

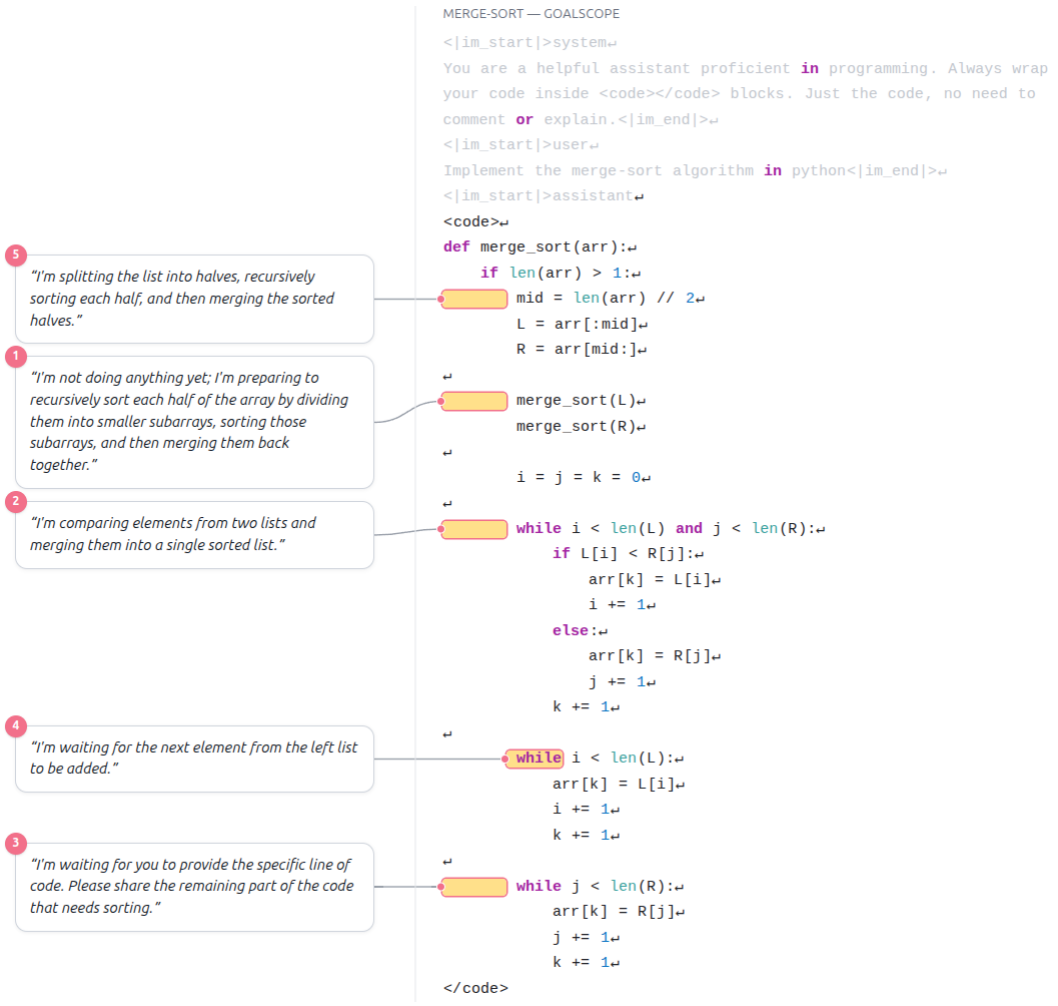


Figure 25: **Reading intent while writing code.** At the marked positions, GoalScope reads descriptions of the operations the model is about to implement: (a) checking for an unmatched man, (b) finding the next free man, and (c) proposing to the next woman on his preference list. These descriptions express an operation rather than a particular token sequence; for example, finding an unmatched man can be implemented with different loop forms. The figure illustrates this distinction and does not itself compare readouts across alternative implementations. Later code is shown for context.



Figure 26: **Reading intent while writing code.** At the marked positions, GoalScope reads descriptions of the operations the model is about to implement: (a) checking for an unmatched man, (b) finding the next free man, and (c) proposing to the next woman on his preference list. These descriptions express an operation rather than a particular token sequence; for example, finding an unmatched man can be implemented with different loop forms. The figure illustrates this distinction and does not itself compare readouts across alternative implementations. Later code is shown for context.

M GENERALIZATION TO NON-CODE SETTINGS

M.1 FEW-SHOT ANTONYM EXAMPLE

The saved exploratory run in `19_goalscope_out_of_domain.ipynb` uses six demonstrations: enter/exit, feasible/infeasible, manned/unmanned, fuzzy/smooth, arise/settle, and far/near. The prompt ends at `Q: mercy` followed by `A:`. Its continuation starts with "cruelty" and then generates further question–answer pairs. The code-localized 122-head set supplies norm-relative additive readouts with $\alpha = .1$, injection layers 35–45, and repeated steering during readout generation.

The probe uses the system instruction "You are a helpful assistant." and the user instruction "You are answering a series of questions that all follow the same hidden rule. In one short phrase, what task are you performing?" This is a domain-adapted probe, not the original code-oriented prompt. Saved token indices refer to the notebook's tokenized prompt plus continuation.

Position	Source token	Readout
57	newline after cruelty	Identifying antonyms.
60	light	Identifying patterns.
70	A before slow	Identifying antonyms.
71	colon before slow	Decoding or interpreting slower actions.

Table 13: Selected qualitative readouts from one saved antonym run, including a vague description and a mismatch. These are not a quantitative evaluation sample.

M.2 OTHER EXPLORATORY DOMAINS

Recipe, travel, and narrative prompts use adapted verbalization questions. These settings require separate decisions about the granularity and correctness of the next step; they are not included in the code results or claimed as a validated general-purpose planning readout. A systematic evaluation remains to be completed.

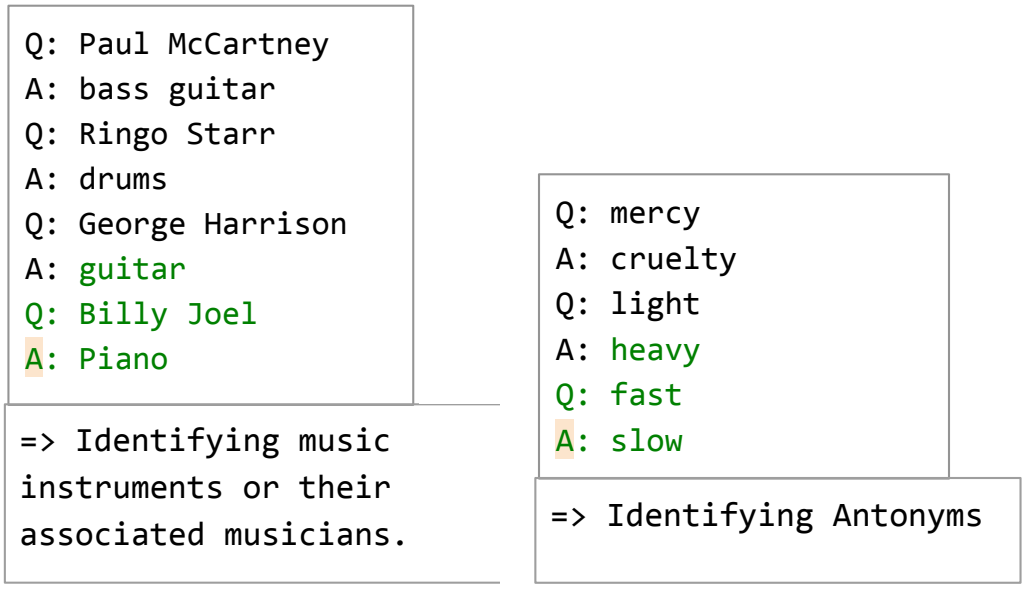


Figure 27: **Reading a relation from few-shot examples.** The slide shows musician–instrument and antonym prompts, with capture positions highlighted on the answer prefix `A`. GoalScope describes the inferred relations. Green text denotes generated continuation shown for context. These selected examples are exploratory and are distinct from the token-indexed antonym run above.

N BOUNDARY SIGNALS: MOVING ON AND STAYING

This supporting line of experiments asks whether signals associated with ending a subgoal can influence when the model advances, independently of the semantic identity read by GoalScope. Prompts permit a predicate to occupy one or several lines, with repeated lines beginning with the current predicate's variable. At a line boundary, the next variable therefore indicates whether the model stays with that predicate or moves to the next one.

Source/destination contrasts are constructed so the source's next variable, the destination's native next variable, and the desired counterfactual variable differ. Head-output patching is optimized to promote the counterfactual continuation, reducing direct copying of a source variable as an explanation. Transfers from completed to incomplete steps can induce moving on; reverse transfers can induce staying or repetition. Variable renaming, cross-language examples, and reversed instruction order probe the reach of this signal.

These observations motivate a distinction between subgoal identity and transitions between subgoals. A next-variable change alone does not show correct implementation of the next predicate, and repeated output under a stay intervention requires separate interpretation. Full continuation scoring and tests of changes to goal readouts are needed before claiming a complete causal dissociation. Boundary experiments are therefore supporting material rather than a premise of the main subgoal-representation account.

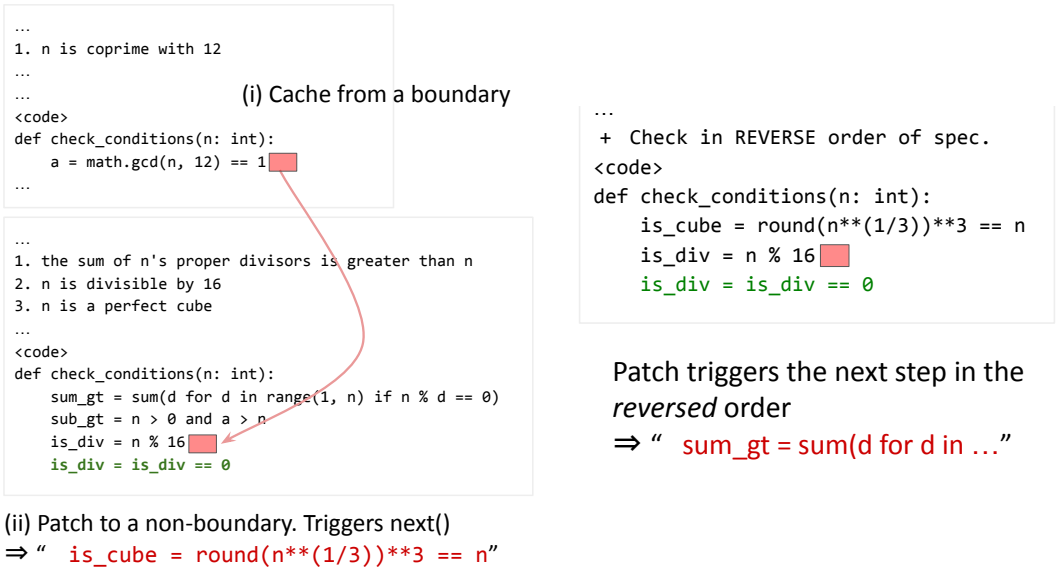


Figure 28: **A boundary intervention under different task orders.** Left: capture at the highlighted completed source expression and patch at the highlighted unfinished divisibility expression in the destination. The green continuation is the native completion of that expression; red text shows the patched run advancing to a new variable. Right: under reversed instruction order, the patched continuation advances to the next task in that order. These are selected examples; advancing to a variable does not certify that the preceding or following predicate is implemented correctly.

O GENERALIZATION ACROSS PROGRAMMING STYLES AND LANGUAGES

The cross-language KV-transfer evaluation is in Appendix H, with aggregate and datatype results in Figures 16 and 17. It covers Python and C++ in both source and destination roles. The free-form examples in Appendix L provide a separate qualitative check beyond the fixed multi-predicate template.

P CHECKING FOR FUNCTIONAL DISSOCIATION

P.1 GOAL HEADS AND IMPULSE HEADS

The saved exploratory comparison uses 122 goal heads and 181 decision/impulse heads in Qwen2.5-Coder-32B. The sets share 36 heads (Jaccard overlap .135). Membership overlap alone does not establish whether the sets perform the same function.

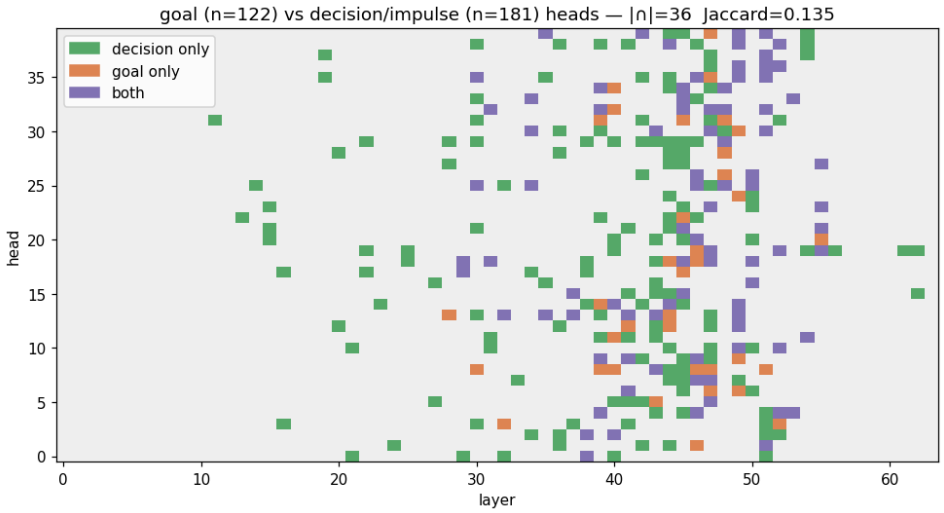


Figure 29: **Goal-head and impulse-head membership.** The saved head sets have partial overlap, concentrated in the later layers. Colors distinguish goal-only, impulse-only, and shared heads.

The accompanying intervention transfers a mean "move-on" signal into either set at the notebook's fixed pivot (relative position −2), using source coefficient 2. The evaluation uses 256 multiline prompts with five conditions and letter-named variables. One metric asks whether the full-vocabulary next-token argmax names the next condition; the other asks whether that token has higher probability than the current condition's token. These are next-token measurements, not GoalScope oracle scores or complete-program correctness.

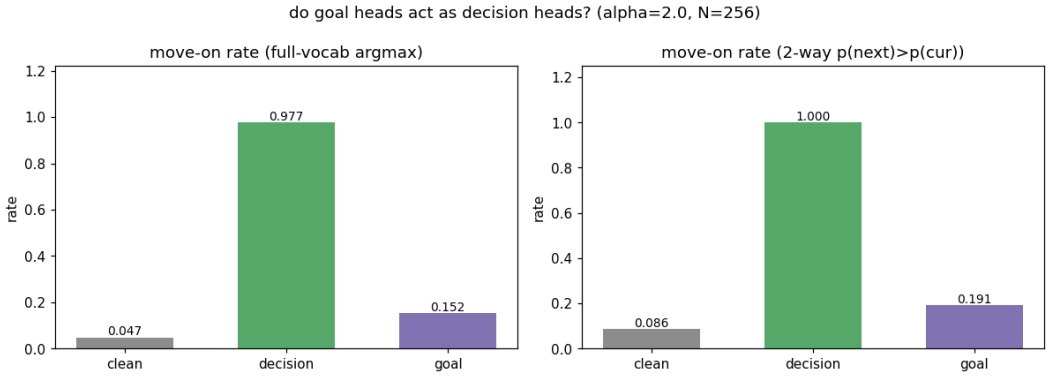


Figure 30: **Transferring a move-on signal through different head sets.** Existing results from `notebooks/clock/11_goal_vs_impulse.ipynb`. Impulse-head intervention increases the move-on rate more than goal-head intervention in this setup. The sets differ in size and depth coverage, so this comparison does not isolate head identity at a matched budget.

Q ROLE OF MLPS

arnab: TODO: path patching experiments: patch + freeze MLPs with dest values. Measure intervention effects.

Placeholder: MLP path-patching results

Compare the goal-head patch with the same patch while restoring MLP outputs to their clean destination values.

Needed: frozen layers and positions, intervention effects, and matched controls.

Figure 31: Reserved for the requested MLP path-patching experiment. A corresponding result figure has not yet been located; this panel reports no measurements.